MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY CODED DECIMAL, DIGITAL, KEY CODE ADK

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary coded decimal counter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.297 failures per million operating hours for class I or 1.025 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ADK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/58-1 for class I and M28787/58-2 for class II. , except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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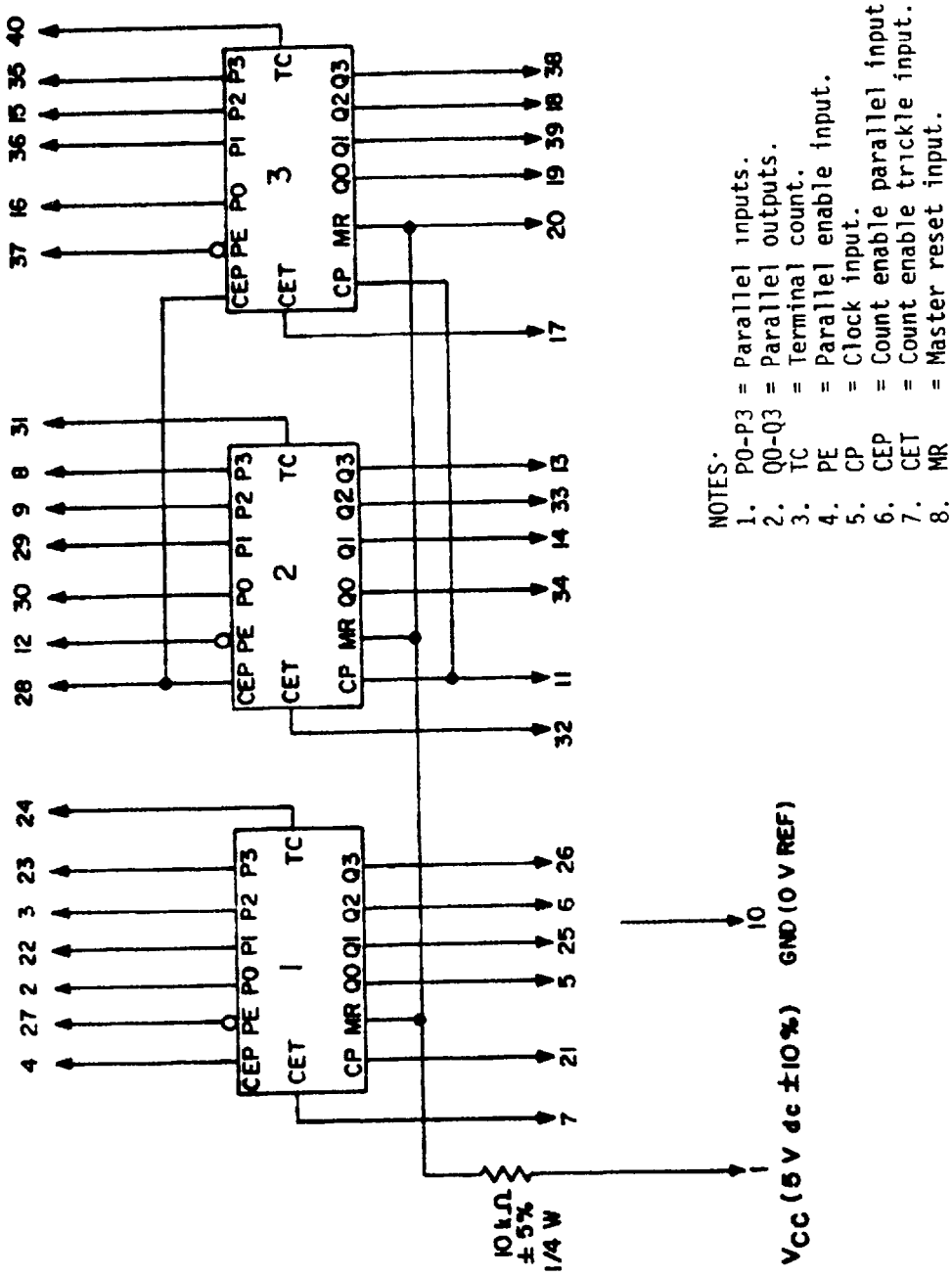


FIGURE 1. ADK module function diagram, binary coded decimal counter.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A15	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
B1-B20	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
B21,B22	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
B23	Input breakdown current	V	S	0	3	0	3	0	3	0	3	0	3	mA
C1-C23	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D	Isolation													
	Initial	VII	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	VII	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
E1 E3	Functional count	VIII	X	← See table VIII →										
F1-F9	Functional (low level)	VIII	X	← See table VIII →										
G1-G15	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H12	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
J1-J3	Input threshold, low level, check	X	X	← See table X →										
K1-K6	Input threshold, high level, check	X	X	← See table X →										
L1	Input current, low level	XII	X	-0.55	-5.77	-0.55	-5.77	-0.55	-5.77	-0.55	-5.77	-0.55	-5.77	mA
L2-L9	Input current, low level	XII	X	-0.10	-3.46	-0.10	-3.46	-0.10	-3.46	-0.10	-3.46	-0.10	-3.46	mA
L10	Input current, low level	XII	X	-0.10	-6.92	-0.10	-6.92	-0.10	-6.92	-0.10	-6.92	-0.10	-6.92	mA
L11	Input current, low level	XII	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
L12-L23	Input current, low level	XII	X	-0.10	-1.12	-0.10	-1.12	-0.10	-1.12	-0.10	-1.12	-0.10	-1.12	mA
M1	Input current, high level	XII	X	-188	-324	-188	-324	-188	-324	-188	-324	-188	-324	µA
M2-M9	Input current, high level	XII	X	0	80	0	80	0	80	0	80	0	80	µA
M10	Input current, high level	XII	X	0	160	0	160	0	160	0	160	0	160	µA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5° C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M11	Input current, high level	XII	X	0	40	0	40	0	40	0	40	0	40	µA
M12-M23	Input current, high level	XII	X	0	27	0	27	0	27	0	27	0	27	µA
N1-N15	Output short circuit current	XIII	X	-20	-80	-20	-80	-20	-80	-20	-80	-20	-80	mA
P1,P2	Power supply current	XIV	S	10	300	10	300	10	300	10	300	10	300	mA
Q1	Input capacitance	XV	Q	3	45	3	45	3	45	3	45	3	45	pF
Q2,Q3	Input capacitance	XV	Q	2	30	2	30	2	30	2	30	2	30	pF
Q4-Q23	Input capacitance	XV	Q	2	15	2	15	2	15	2	15	2	15	pF
R	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
S1-S15	Transition time (t_{THL})	XVII	S	2	20	2	20	2	25	2	20	2	25	ns
T1-T15	Transition time (t_{TLH})	XVII	S	5	35	5	35	5	40	5	35	5	40	ns
U1-U3	Propagation delay time (t_{PHL})	XVII	S	2	45	2	45	2	50	2	45	2	50	ns
U4-U15	Propagation delay time (t_{PHL})	XVII	S	2	40	2	40	2	50	2	40	2	50	ns
V1-V3	Propagation delay time (t_{PLH})	XVII	S	4	65	4	65	4	70	4	65	4	70	ns
V4-V15	Propagation delay time (t_{PLH})	XVII	S	4	45	4	45	4	50	4	45	4	50	ns
W1-W15	Propagation delay time (t_{PHL}) and minimum MR pulse width and recovery time check	XVIII	S	5	65	5	65	5	70	5	65	5	70	ns
X1-X3	Propagation delay time (t_{PHL})	XIX	S	2	45	2	45	2	50	2	45	2	50	ns
Y1-Y3	Propagation delay time (t_{PLH})	XIX	S	2	45	2	45	2	50	2	45	2	50	ns
Z1-Z3	Maximum counting frequency (minimum clock pulse width) check (PW # 1)	XX	S	85	115	85	115	85	115	85	115	85	115	ns
Z4-Z6	Maximum counting frequency (minimum clock pulse width) check (PW # 2)	XX	S	385	415	385	415	385	415	385	415	385	415	ns
AA1-AA6	Setup and hold time (CET AND CEP) (measuring output PRP) check	XXI	S	1.9	2.1	1.9	2.1	1.9	2.1	1.9	2.1	1.9	2.1	µs

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
BB1-BB12	Setup and hold time (P) (measuring t_{PLH}), high level, check	XXII	S	5	45	5	45	5	50	5	45	5	50	ns
BB13-BB15	Setup and hold time (PE) (measuring t_{PLH}), low level, check	XXII	S	5	45	5	45	5	50	5	45	5	50	ns
BB16-BB27	Setup and hold time (P) (measuring t_{PHL}), low level, check	XXII	S	5	45	5	45	5	50	5	45	5	50	ns
CC1-CC15	Output crosstalk, high level	XXIII	Q	0	400	0	400	0	400	0	400	0	400	mV
DD1-DD15	Output crosstalk, low level	XXIII	Q	0	800	0	800	0	800	0	800	0	800	mV
DD16-DD30	Output crosstalk, duration	XXIII	Q	0	400	0	400	0	400	0	400	0	400	mV
EE	Power supply filter capacitance	XXIV	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, B, or D are part of a sequence of tests, the order of testing shall be as follows. A, B, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through EE in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C, E through P, and S through EE shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition EE, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C, E through P, and R through EE shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C and E through EE shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/58-1 supersedes part number M28787/58. The two parts are interchangeable.

6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Power dissipation (outputs not loaded)	1.65 max	1.65 max	W
Supply voltage	5 ±10%	5 ±10%	V dc

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Input voltage				
Low level		-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level		2 min, V_{CC} max	2 min, V_{CC} max	V
Input current				
CP-1, CET, PE, CEP2, CEP3				
Low level	(V_{IL} = 0 V)	-0.1 min, -3.46 max	-0.1 min, -3.46 max	mA
	(V_{IL} = 0.4 V)	-0.1 min, -3.2 max	-0.1 min, -3.2 max	mA
High level	(V_{IH} = 2.4 V)	0 min, 80 max	0 min, 80 max	µA
	(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
MR				
Low level	(V_{IL} = 0.0 V)	-0.55 min, -5.77 max	-0.55 min, -5.77 max	mA
	(V_{IL} = 0.4 V)	-0.55 min, -5.33 max	-0.55 min, -5.33 max	mA
High level	(V_{IH} = 2.4 V)	-188 min, -324 max	-188 min, -324 max	µA
	(V_{IH} = 5.5 V)	3 max	3 max	mA
P0-P3				
Low level	(V_{IL} = 0.0 V)	-0.1 min, -1.12 max	-0.1 min, -1.12 max	mA
	(V_{IL} = 0.4 V)	-0.1 min, -1.07 max	-0.1 min, -1.07 max	mA
High level	(V_{IH} = 2.4 V)	0 min, 27 max	0 min, 27 max	µA
	(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
CEP-1				
Low level	(V_{IL} = 0.0 V)	-0.1 min, -1.73 max	-0.1 min, -1.73 max	mA
	(V_{IL} = 0.4 V)	-0.1 min, -1.6 max	-0.1 min, -1.6 max	mA
High level	(V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
	(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
CP-2 and CP-3				
Low level	(V_{IL} = 0.0 V)	-0.1 min, -6.92 max	-0.1 min, -6.92 max	mA
	(V_{IL} = 0.4 V)	-0.1 min, -6.4 max	-0.1 min, -6.4 max	mA
High level	(V_{IH} = 2.4 V)	0 min, 160 max	0 min, 160 max	µA
	(V_{IH} = 5.5 V)	0 min, 2 max	0 min, 2 max	mA
Output voltage				
Low level		0.0 min, 0.4 max	0.0 min, 0.4 max	V
High level		2.4 min, 4.5 max	2.4 min, 4.5 max	V
Output drive current maximum capability				
Low level	(V_{OL} = 0.4 V)	12.8 max	12.8 max	mA
High level	(V_{OH} = 2.4 V)	-640 max	-640 max	µA
Clock input				
Frequency		20 max	20 max	MHz
Pulse width (high level)		20 min	20 min	ns
Propagation delay time				
t_{PLH}				
CP to TC		4 min, 70 max	4 min, 70 max	ns
CP to Q		4 min, 50 max	4 min, 50 max	ns
CET to TC		2 min, 50 max	2 min, 50 max	ns
t_{PHL}				
CP to TC		2 min, 50 max	2 min, 50 max	ns
CP to Q		2 min, 50 max	2 min, 50 max	ns
CET to TC		2 min, 50 max	2 min, 50 max	ns
MR to Q		5 min, 70 max	5 min, 70 max	ns
Setup time				
CET and CEP		30 min	30 min	ns
P0-P3		38 min	38 min	ns
PE		45 min	45 min	ns

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Hold time			
CET and CEP	0 min	0 min	ns
PO-P3	0 min	0 min	ns
PE	0 min	0 min	ns
Reset pulse width	35 min	35 min	ns
Reset recovery time	20 min	20 min	ns

* 6.6 Module function. This module contains three 4 bit synchronous BCD counters with independent inputs and outputs with the exception that circuits 2 and 3 have common clocks (CP) and counter enable parallel inputs (CEP). All counters have a common master reset (MR) input.

6.6.1 Input to output relationship. Each counter has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 5/

Synchronous inputs before clock pulse							Required Condition of asynchronous inputs	Outputs after clock pulse goes high				
CEP	CET	PE	P3	P2	P1	P0	MR	TC	Q3	Q2	Q1	Q0
L	X	H	D3	D2	D1	D0	H	X	NC	NC	NC	NC
X	L	H	D3	D2	D1	D0	H	L	NC	NC	NC	NC
X	X	L	D3	D2	D1	D0	H	L	D3	D2	D1	D0
X	X	L	H	H	H	H	H	L	H	H	H	H
X	X	L	L	L	L	L	H	L	L	L	L	L
H	H	H	X	X	X	X	H	L	L	L	L	H
H	H	H	X	X	X	X	H	L	L	L	H	L
H	H	H	X	X	X	X	H	L	L	L	H	H
H	H	H	X	Y	Y	Y	H	L	L	H	L	L
H	H	H	X	X	X	X	H	L	L	H	L	H
H	H	H	X	X	X	X	H	L	L	H	H	L
H	H	H	X	X	X	X	H	L	L	H	H	H
H	H	H	X	X	X	X	H	L	H	L	L	L
H	H	H	X	X	X	X	H	H	H	L	L	H

1/ NC = No change from previous state.
2/ D0-D3 = Known input and output data (either high level or low level)
3/ X = Unspecified inputs or outputs (either high level or low level).
4/ H = High level
5/ L = Low level

6.6.1.1 Asynchronous master reset. The asynchronous master reset (MR) input sets the counter outputs to low level independent of any other input.

6.6.1.2 Functional description. The ADK module contains three BCD decade counters which are fully synchronous with the clock pulse. Each counter contains four master or slave flip-flops which are driven, in parallel, by the clock through a clock buffer. The master is inhibited from further change during the low-to-high transition of the clock. Data is transferred from the masters to the slaves and reflected at the outputs after the masters are inhibited. When the clock is high, the masters remain inhibited and the master or slave data path remains established. During the clock high-to-low transition, the slave is inhibited from further change, followed by the enabling of the masters for acceptance of new data from the counting logic or the parallel entry logic.

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6.6.1.3 Control inputs. There are three control inputs. Parallel enable (PE), count enable parallel (CEP), and count enable trickle (CET). These control inputs select the mode of operation as shown in table III. When the conditions for counting are satisfied, that is, CEP, CET, and PE are high, the rising edge of the clock will change the counters to the state of the count sequence.

TABLE III. Mode selection. 1/

PE	CEP	CET	Mode
L	L	L	Preset
L	L	H	Preset
L	H	L	Preset
L	H	H	Preset
H	L	L	No change
H	L	H	No change
H	H	L	No change
H	H	H	Count

1/ MR = High.

6.6.1.4 Counter setting. The counter can be synchronously preset from the four parallel inputs (P0-P3), when PE is low. When the PE and clock are low, each master of the flip-flops is connected to the appropriate parallel input (P0-P3) and the slaves (outputs) are steady in their previous state. When the clock goes high, this information is transferred to the slaves and reflected at the outputs. The PE overrides both count enables presetting the counter low.

6.6.1.5 State of terminal count. Terminal count is high when the counter is at a terminal count (state 9) and the CET is high.

6.6.1.6 Overriding provisions. The asynchronous master reset, common to all counters, overrides all other inputs resetting each output low.

6.6.1.7 Special conditions. If PE goes high while the clock is low and count enables are not active (low) during the low portion of the clock, the low-to-high clock transition will change (Q0-Q3 to the P0-P3 data that existed as the setup time before the rising edge of PE.

a. If the count enables are active (high) during the portion of the clock low period, but PE is high during the entire clock low period, the subsequent low-to-high transition will change Q0-Q3 to the next count value.

b. If PE goes high while the clock is low, but the count enables are active (high) during some portion of the remaining clock low period, the counter will perform a mixture of counting and loading. On the low-to-high clock transition, outputs Q0-Q3 will change as the count sequence or the loading requires. Only the outputs that would not change in the count sequence and are also reloaded with their present value stay constant.

6.6.1.8 Common CEP and master reset. Two of the counters in the module have a common CEP and all three have a common master reset.

6.6.1.9 High-speed lookahead technique. Multistage synchronous counting at high speeds is made possible with a high-speed lookahead technique utilizing the count enable and terminal count logic.

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6.6.1.10 TC (terminal count) = CET · Q0 · $\overline{Q1}$ · $\overline{Q2}$ · Q3. The TC output is asynchronous, therefore, the TC output should never be used outside the counting circuit as a synchronous output unless resynchronizing circuitry is added to compensate for the delay differences between the TC and Q outputs. When connected as a multistage counter (TC output connected to CET input), the delay t_{PHL} from CP to TC of a counter may exceed the delay t_{PLH} from CP to Q0 of the following (adjacent) counter. Under the conditions of the first circuit changing from count 9_{10} to 0 and the following (MSB) circuit changing from count 8_{10} to 9_{10}, the TC output of that MSB counter could generate an illegal logic "high" pulse.

6.6.2 Parallel outputs. Paralleling of outputs is not permissible.

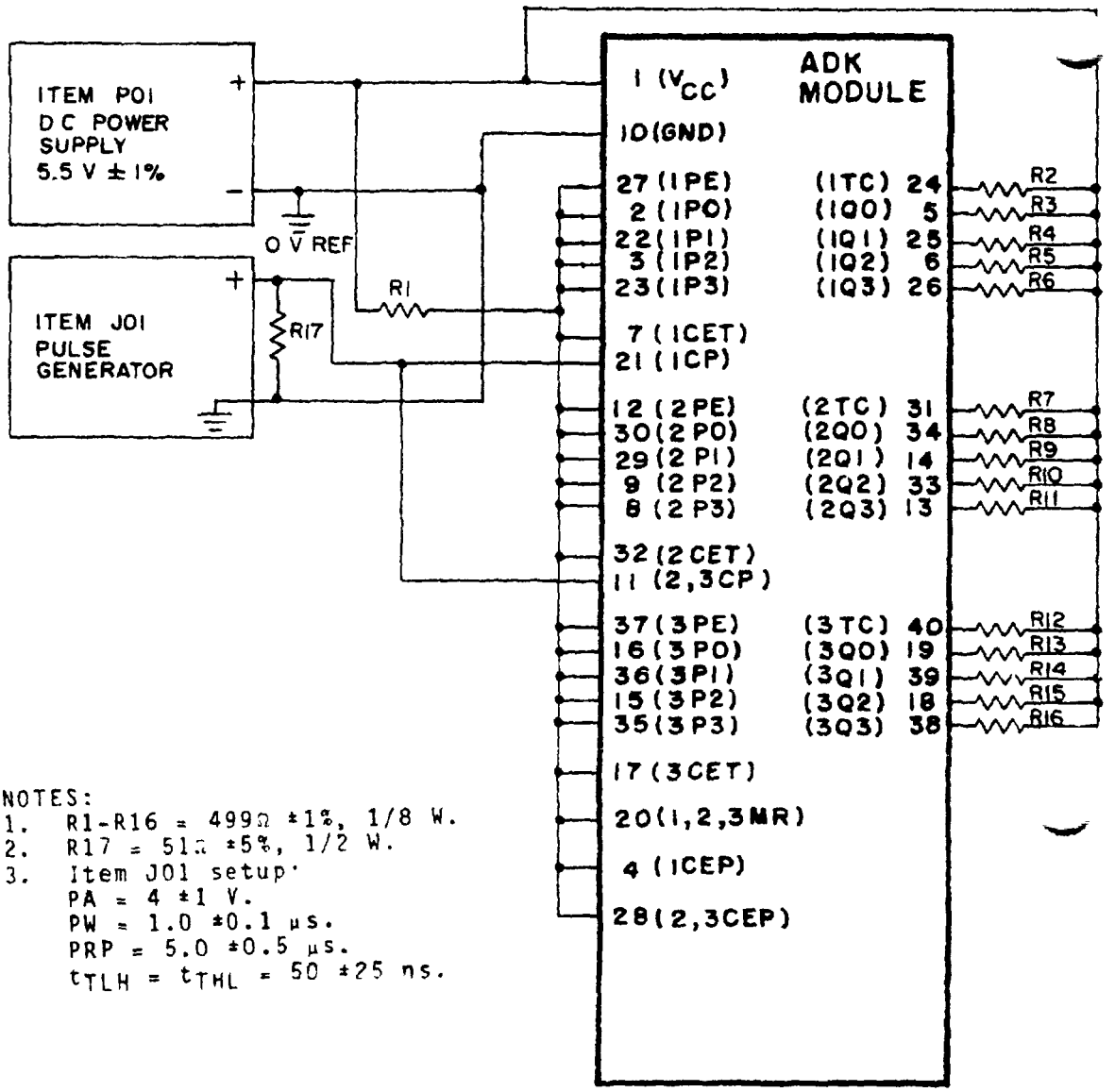
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent
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Preparing Activity:
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(Project 5963-N468)

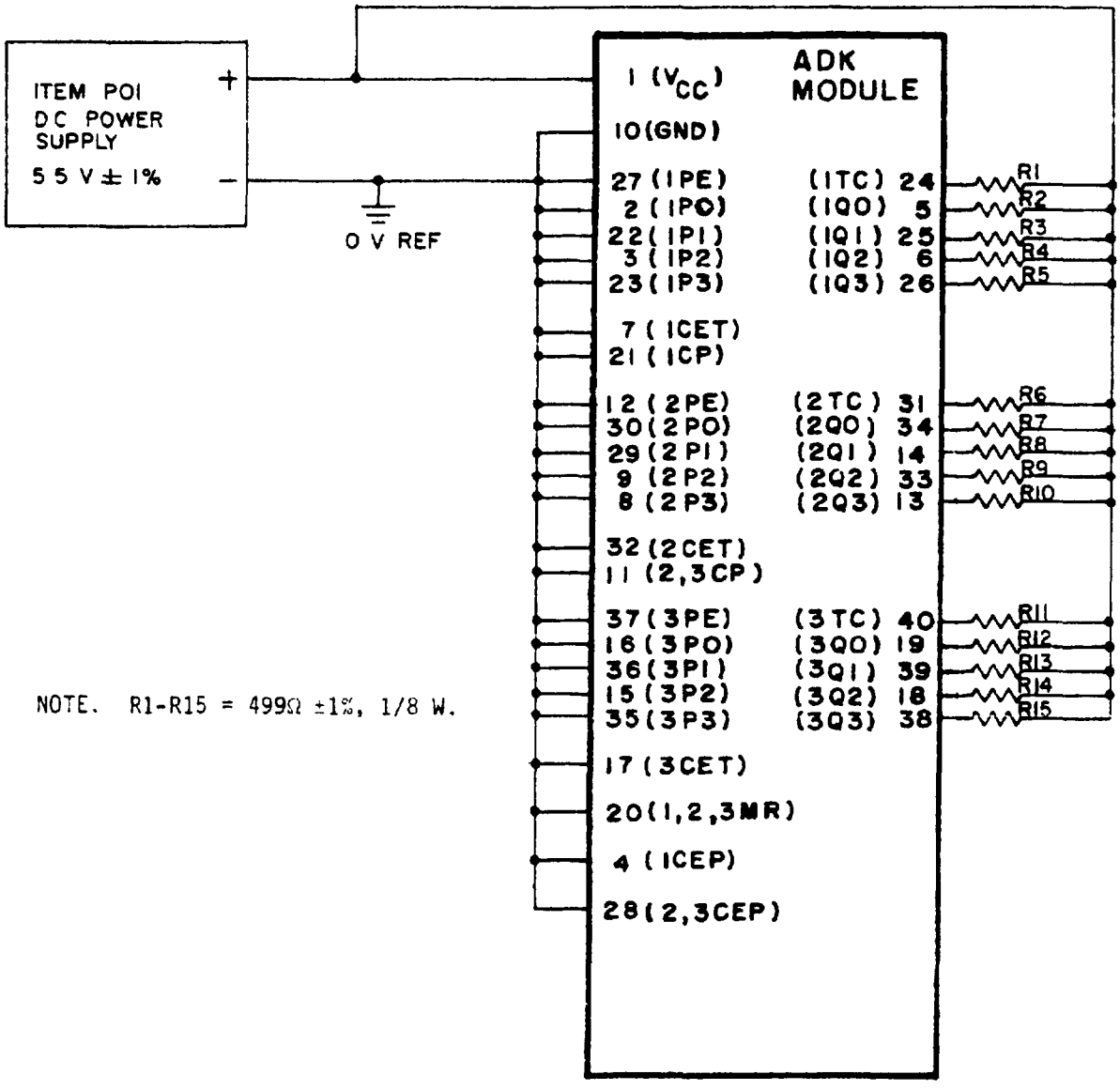
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NOTES:
1. R1-R16 = 499Ω ±1%, 1/8 W.
2. R17 = 51Ω ±5%, 1/2 W.
3. Item J01 setup:
   PA = 4 ±1 V.
   PW = 1.0 ±0.1 µs.
   PRP = 5.0 ±0.5 µs.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.

* FIGURE 2 Operating temperature and life test circuit.

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NOTE. R1-R15 = 499Ω ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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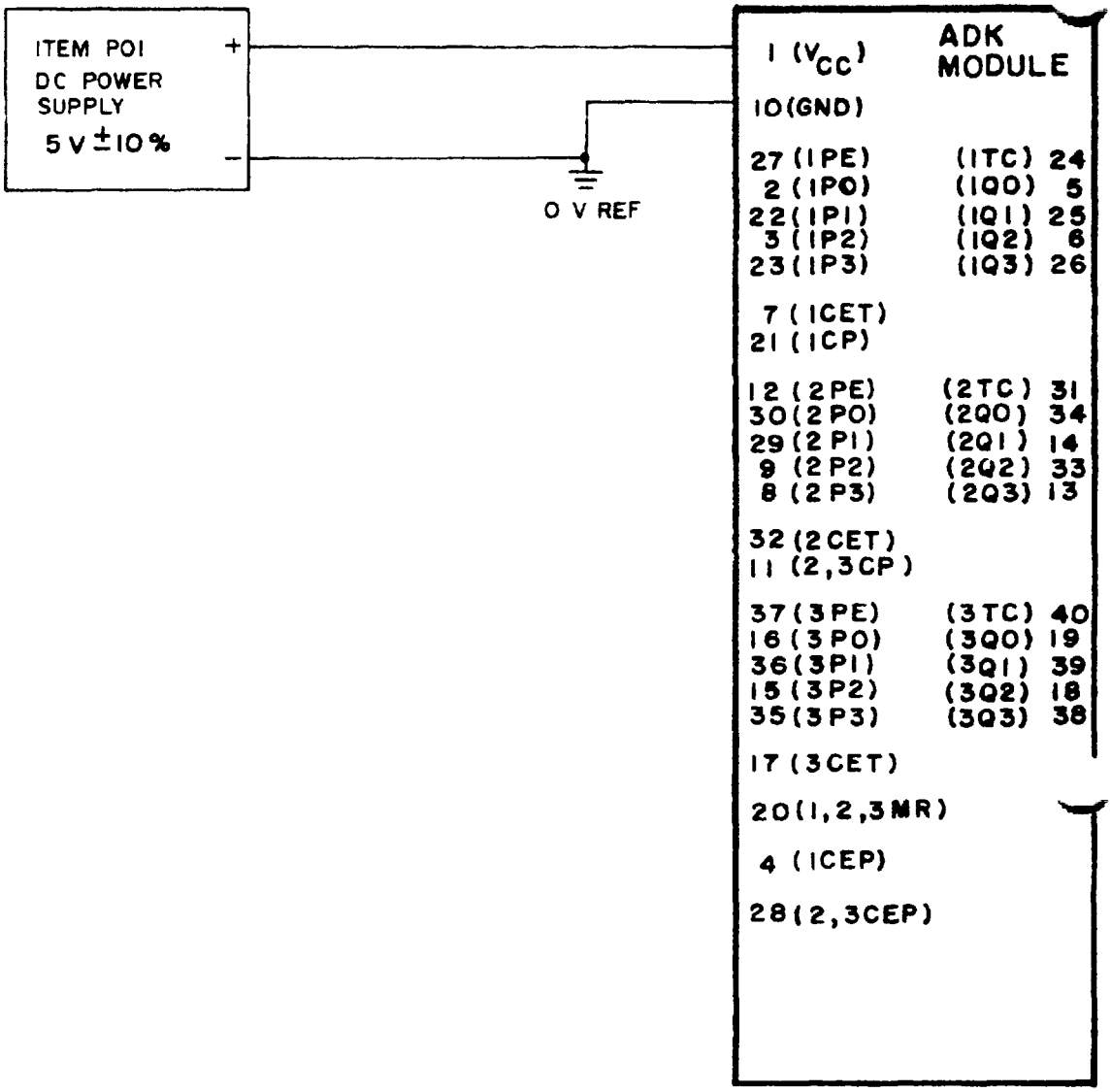


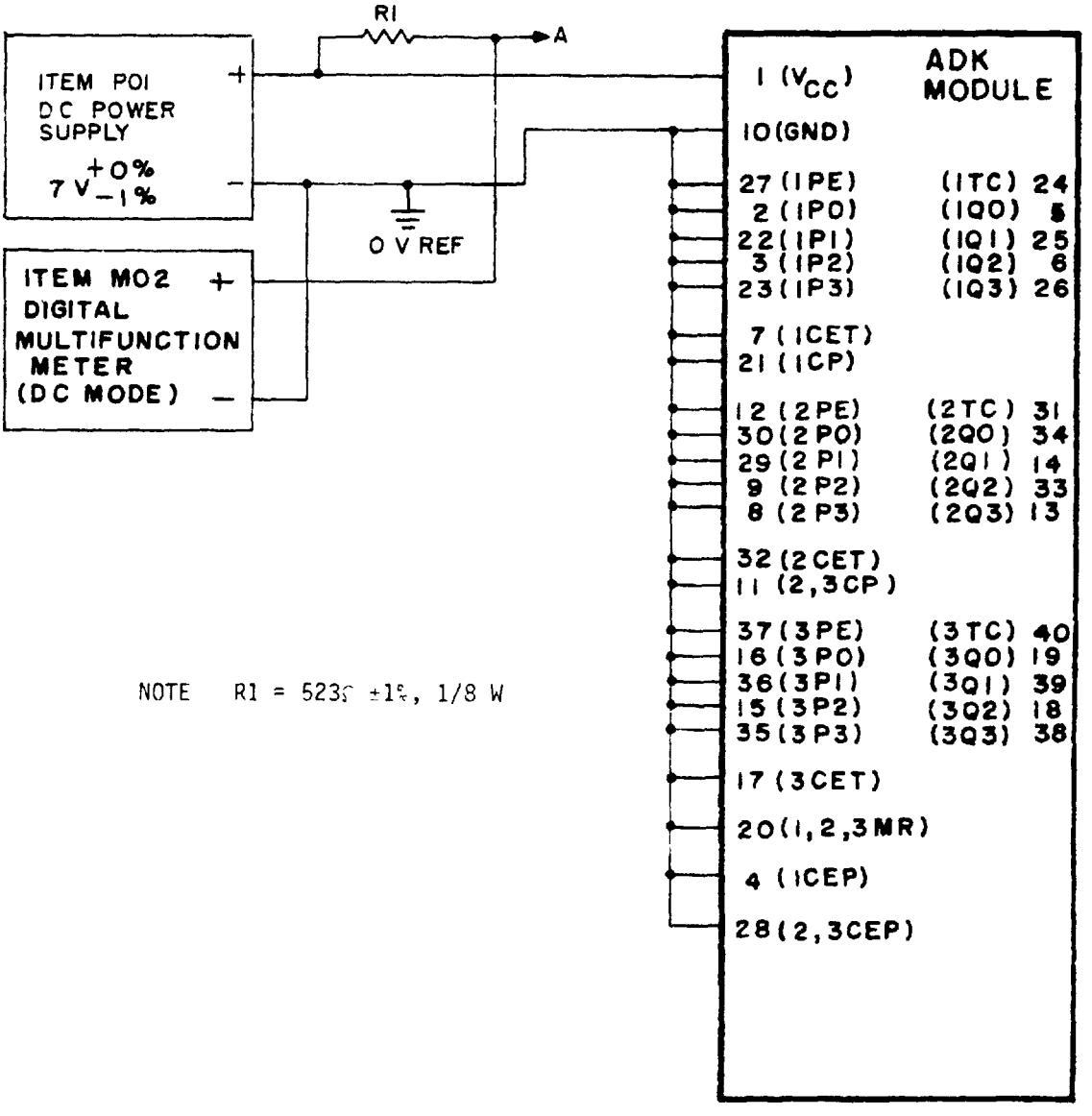
FIGURE 4. Vibration test circuit.

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*TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
A1	1Q3 V_{OUT} (low loaded)	26	10	800	mV	5
A2	1Q2 "	6	"	"	"	"
A3	1Q1 "	25	"	"	"	"
A4	1Q0 "	5	"	"	"	"
A5	1TC "	24	"	"	"	"
A6	2Q3 "	13	"	"	"	"
A7	2Q2 "	33	"	"	"	"
A8	2Q1 "	14	"	"	"	"
A9	2Q0 "	34	"	"	"	"
A10	2TC "	31	"	"	"	"
A11	3Q3 "	38	"	"	"	"
A12	3Q2 "	18	"	"	"	"
A13	3Q1 "	39	"	"	"	"
A14	3Q0 "	19	"	"	"	"
A15	3TC "	40	"	"	"	"

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NOTE R1 = 523Ω ±1%, 1/8 W

*FIGURE 5. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1CEP Input breakdown current	4	2,3,7,20,21,22,23,27	0	1	mA	6
B2	1CP "	21	2,3,4,7,20,22,23,27	"	"	"	"
B3	1CET "	7	2,3,4,20,21,22,23,27	"	"	"	"
B4	1PE "	27	2,3,4,7,20,21,22,23	"	"	"	"
B5	2PE "	12	8,9,11,20,28,29,30,32,37	"	"	"	"
B6	2CET "	32	8,9,11,12,20,28,29,30,37	"	"	"	"
B7	3CET "	17	11,15,16,20,28,35,36,37	"	"	"	"
B8	1P0 "	2	3,4,7,20,21,22,23	"	"	"	"
B9	1P1 "	22	2,3,4,7,20,21,23	"	"	"	"
B10	1P2 "	3	2,4,7,20,21,22,23	"	"	"	"
B11	1P3 "	23	2,3,4,7,20,21,22	"	"	"	"
B12	2P0 "	30	8,9,11,12,20,28,29,32	"	"	"	"
B13	2P1 "	29	8,9,11,20,28,30,32	"	"	"	"
B14	2P2 "	9	8,11,20,28,29,30,32	"	"	"	"
B15	2P3 "	8	9,11,20,28,29,30,32	"	"	"	"
B16	3P3 "	35	11,15,16,17,20,28,36	"	"	"	"
B17	3P2 "	15	11,16,17,20,28,35,36	"	"	"	"
B18	3P1 "	36	11.15,16,17,20,28,35	"	"	"	"
B19	3P0 "	16	11 15,17,20,28,35,36	"	"	"	"
B20	3PE "	37	11.15,16,17,20,28,35,36	"	"	"	"
B21	2,3 CEP "	28	8,9,11,12,15,16,17,20, 29,30,32,35,36,37	"	2	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B22	2,3 CP	Input breakdown current	11	8,9,12,15,16,17,20,28, 29,30,32,35,36,37	0	2	mA	6
B23	1,2,3 MR	Input breakdown current	20	2,3,4,7,8,9,11,12,15,16, 17,21,22,23,27,28,29,30, 32,35,36,37	0	3	mA	6

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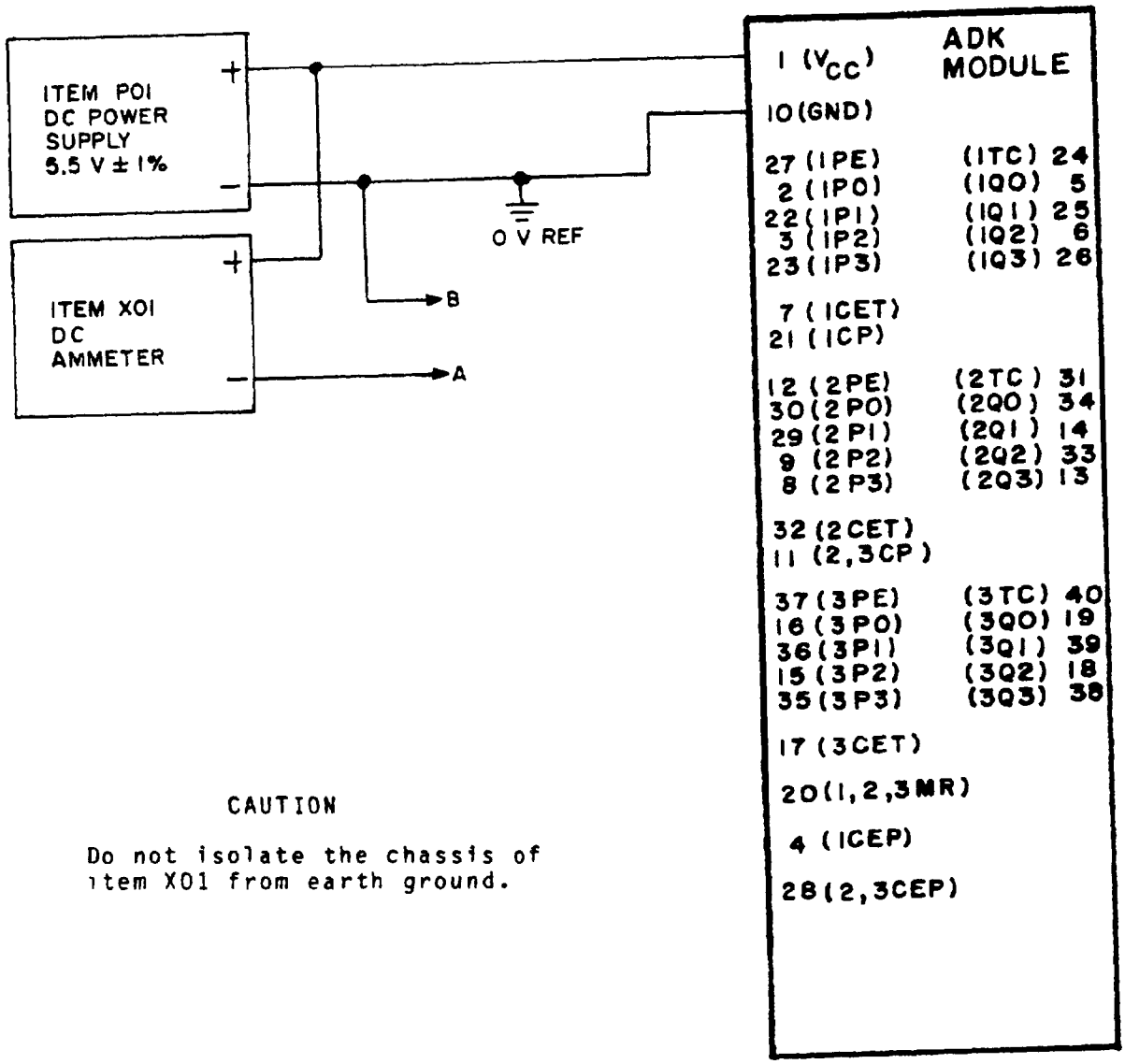


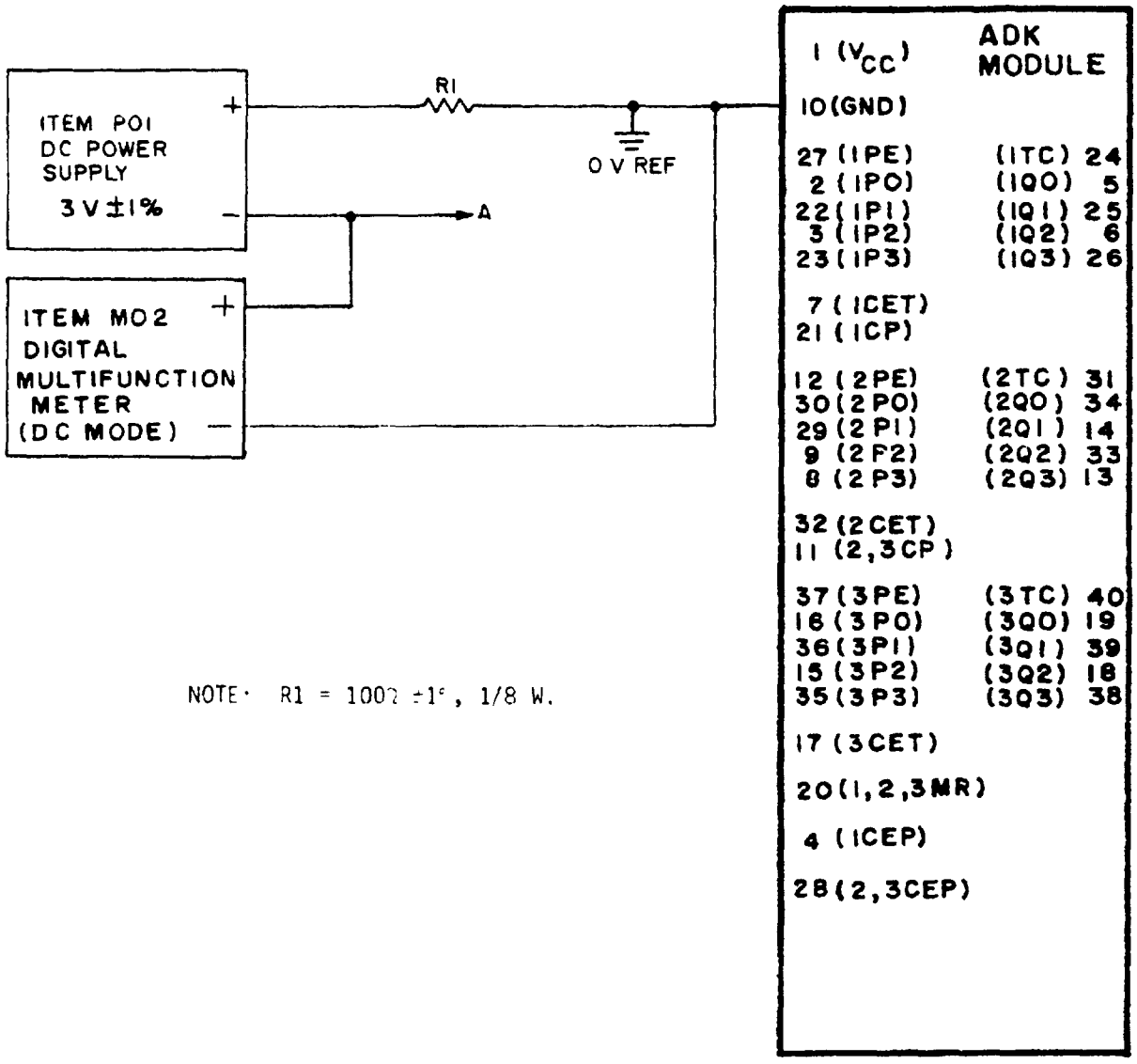
FIGURE 6. Input breakdown current test circuit for table V.

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TABLE VI Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	1,2,3 MR Input clamp diode voltage	20	-0.4	-1.5	V	7
C2	1CP "	21	"	"	"	"
C3	1PO "	2	"	"	"	"
C4	1P1 "	22	"	"	"	"
C5	1P2 "	3	"	"	"	"
C6	1P3 "	23	"	"	"	"
C7	1CET "	7	"	"	"	"
C8	1PE "	27	"	"	"	"
C9	1CEP "	4	"	"	"	"
C10	2CP "	11	"	"	"	"
C11	2PO "	30	"	"	"	"
C12	2P1 "	29	"	"	"	"
C13	2P2 "	9	"	"	"	"
C14	2CET "	32	"	"	"	"
C15	2PE "	12	"	"	"	"
C16	2P3 "	8	"	"	"	"
C17	3CET "	17	"	"	"	"
C18	3PE "	37	"	"	"	"
C19	3P3-3 "	35	"	"	"	"
C20	3P2-3 "	15	"	"	"	"
C21	3P1-3 "	36	"	"	"	"
C22	3PO-3 "	16	"	"	"	"
C23	2,3 CEP "	28	"	"	"	"

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NOTE: R1 = 100Ω ±1%, 1/8 W.

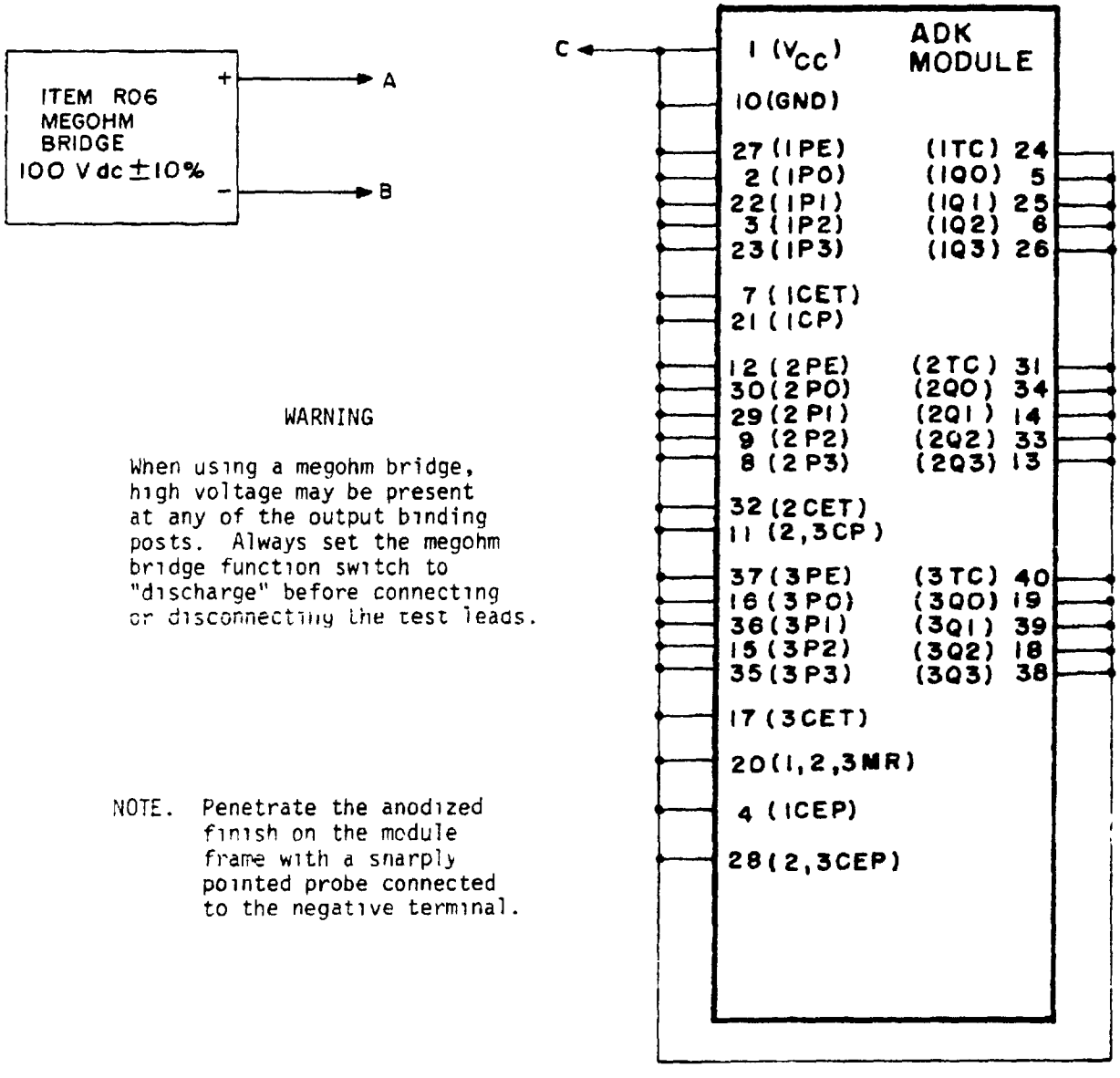
*FIGURE 7. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D	Isolation	C	Frame	10	∞	mΩ	8

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE. Penetrate the anodized finish on the module frame with a snarply pointed probe connected to the negative terminal.

FIGURE 8 Isolation test circuit for table VII

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*TABLE VIII. Functional (count and low-level) test.

Test condition	Parameter	Connect								Initial limits 25°C ±5°C ambient		Unit	Output logic states					Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		TC	Q3	Q2	Q1	Q0	
E1	Unit 1 Functional Count	Open	2,3, 22,23, G	4,7, 27	Open	Open	21	B	20	See figure 9, notes 5 and 6			See table XI					9
E2	Unit 2 "	"	8,9, 29,30, G	12, 28,32	"		11	"	"	"			"					"
E3	Unit 3 "	"	15,16, 35,36, G	17, 28,37	"		11	"	"	"			"					"
F1	1CET Functional, low level	2,23,4 27	3,22,G	Open	7	Open	21	"	"	See figure 9, notes 5 and 7			L	L	L	L	L	"
F2	1PE "	2,23,4	3,22,G	"	7,27	24	21	"	"	"			L	H	L	L	H	"
F3	2CET "	8,12,28, 30	9,29,G	"	32	Open	11	"	"	"			L	L	L	L	L	"
F4	2PE "	8,30,28,	9,29,G	"	12,32	31	"	"	"	"			L	H	L	L	H	"
F5	3CET "	37,16, 35,28	15,36,G	"	17	Open	"	"	"	"			L	L	L	L	L	"
F6	3PE "	16,35, 28	15,36,G	"	17,37	40	"	"	"	"			L	H	L	L	H	"
F7	1CEP "	2,3,7, 22,23, 27	G	"	4	Open	21	"	"	"			L	L	L	L	L	"
F8	2CEP "	8,9,29, 30,32, 12	"	"	28	"	11	"	"	"			L	L	L	L	L	"
F9	3CEP "	15,16, 17,35, 36,37	"	"	29	"	11	"	"	"			L	L	L	L	L	"

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*TABLE IX. Output voltage (high-level and low-level) test.

Test condition	Parameter	Connect								Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
G1	1TC Output voltage, high level	4,7	27,G	2,23	3,22	24	21	B	20	2.4	4.5	V	9
G2	1Q3 "	Open	27,4, 7,G	23	2,22,3	26	"	"	"	"	"	"	"
G3	1Q2 "	"	"	3	2,22, 23	6	"	"	"	"	"	"	"
G4	1Q1 "	"	"	22	2,3,23	25	"	"	"	"	"	"	"
G5	1Q0 "	"	"	2	3,22, 23	5	"	"	"	"	"	"	"
G6	2TC "	28,32	12,G	8,30	9,29	31	11	"	"	"	"	"	"
G7	2Q3 "	Open	12,28, 32,G	8	9,29, 30	13	"	"	"	"	"	"	"
G8	2Q2 "	"	"	9	8,29, 30	33	"	"	"	"	"	"	"
G9	2Q1 "	"	"	29	8,9,30	14	"	"	"	"	"	"	"
G10	2Q0 "	"	"	30	8,9,29	34	"	"	"	"	"	"	"
G11	3TC "	28,17	37,G	16,35	15,36	40	"	"	"	"	"	"	"
G12	3Q3 "	Open	17,28, 37,G	35	15,16, 36	38	"	"	"	"	"	"	"
G13	3Q2 "	"	"	15	16,35, 36	18	"	"	"	"	"	"	"
G14	3Q1 "	"	"	36	15,16, 35	39	"	"	"	"	"	"	"
G15	3Q0 "	"	"	16	15,35, 36	19	"	"	"	"	"	"	"
H1	1Q3 Output voltage, low level	"	4,7, 27,G	2,3, 22	23	26	21	"	"	0	400	mV	"
H2	1Q2 Output voltage, low level	"	4,7, 27,G	2,22, 23	3	6	21	"	"	0	400	mV	"

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*TABLE IX. Output voltage (high-level and low-level) test - Continued.

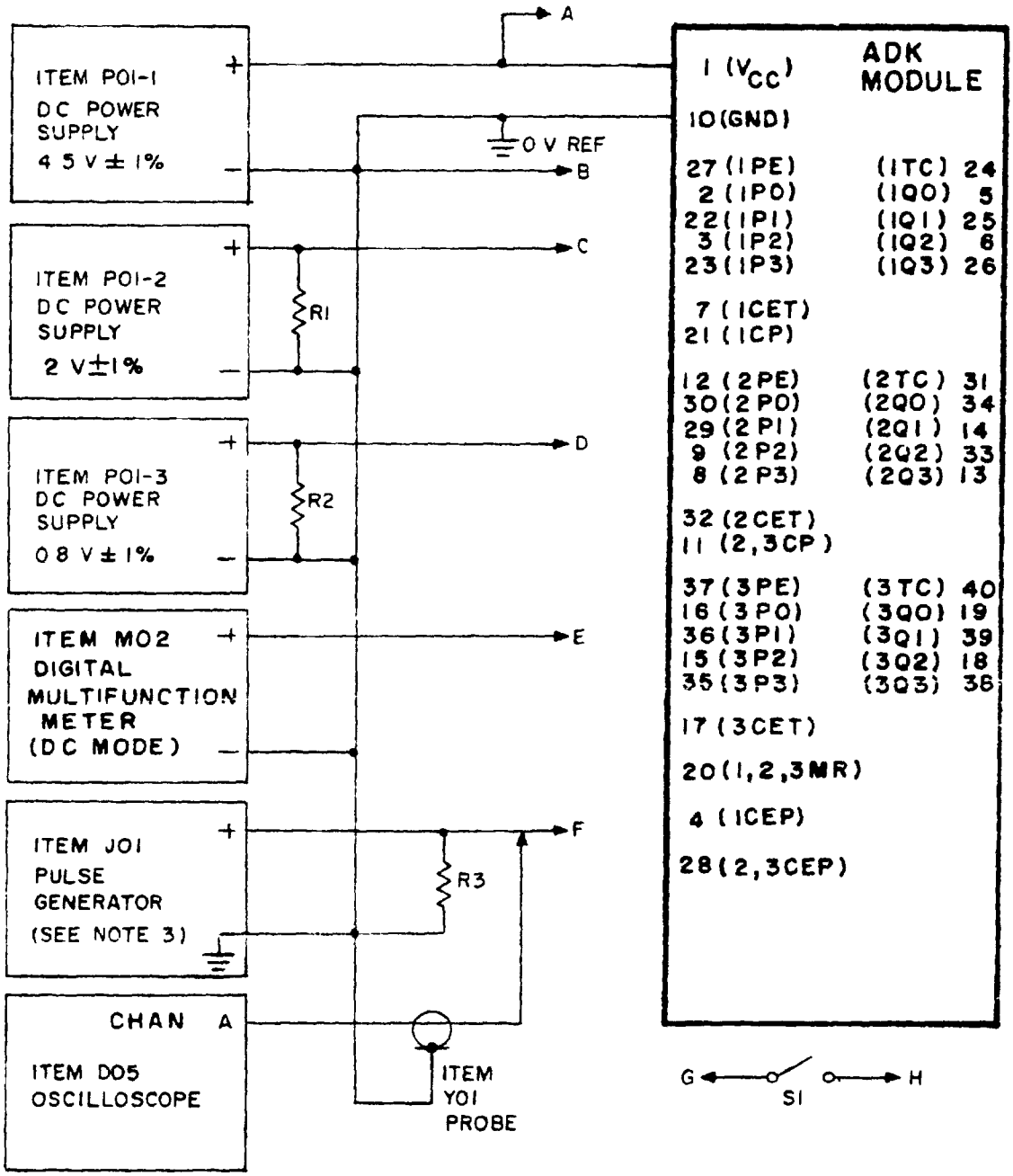
Test condition	Parameter	Connect								Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
H3	1Q1 Output voltage, low level	Open	4,7, 27,G	2,3, 23	22	25	21	B	20	0	400	mV	9
H4	1Q0 "	"	4,7, 27,G	3,22, 23	2	5	21	"	"	"	"	"	"
H5	2Q3 "	"	12,28, 32,G	9,29, 30	8	13	11	"	"	"	"	"	"
H6	2Q2 "	"	"	8,29, 30	9	33	"	"	"	"	"	"	"
H7	2Q1 "	"	"	8,9, 30	29	14	"	"	"	"	"	"	"
H8	2Q0 "	"	"	8,9, 29	30	34	"	"	"	"	"	"	"
H9	3Q3 "	"	17,28, 37,G	15,16 36	35	38	"	"	"	"	"	"	"
H10	3Q2 "	"	"	16,35 36	15	18	"	"	"	"	"	"	"
H11	3Q1 "	"	"	15,16 35	36	39	"	"	"	"	"	"	"
H12	3Q0 "	"	"	15,35 36	16	19	"	"	"	"	"	"	"

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*TABLE X. Input threshold (low-level and high-level) check test.

Test condition	Parameter	Connect								Initial limits 25°C ±5°C ambient		Unit	Output logic states					Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		TC	Q3	Q2	Q1	Q0	
J1	1MR Input threshold, low level, check	2,3,4, 7,22,23	27	Open	G	Open	20	D	21	See figure 9, notes 5 and 9			L	L	L	L	L	9
J2	2MR "	8,9,28, 29,30, 32	12	"	"	"	"	"	11	"			L	L	L	L	L	"
J3	3MR "	15,16, 17,28, 35,36	37	"	"	"	"	"	11	"			L	L	L	L	L	"
K1	1CET Input threshold, high level, check	2,4,23	3,22, 27,G	7	Open	"	21	B	20	See figure 9, notes 5 and 10			H	H	L	L	H	"
K2	2CET "	8,28,30	9,12, 29,G	32	"	"	11	"	"	"			H	H	L	L	H	"
K3	3CET "	16,28, 35	15,36, 37,G	17	"	"	11	"	"	"			H	H	L	L	H	"
K4	1CEP "	2,7,23,	3,22, 27,G	4	"	"	21	"	"	"			H	H	L	L	H	"
K5	2CEP "	8,30,32	9,12, 29,G	28	"	"	11	"	"	"			H	H	L	L	H	"
K6	3CEP "	16,17, 35	15,36, 37,G	28	"	"	11	"	"	"			H	H	L	L	H	"

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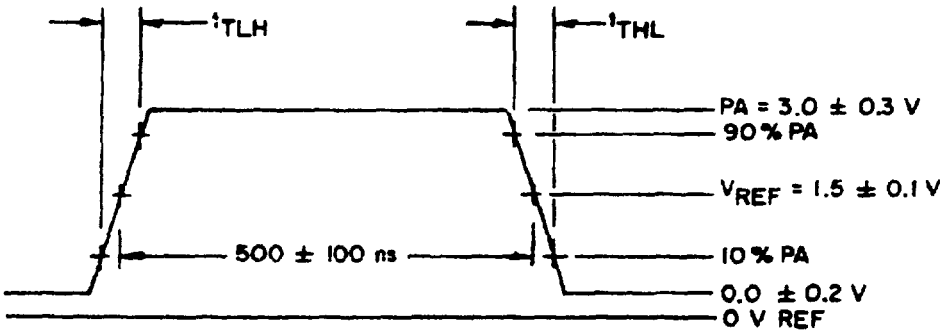


* FIGURE 9. Functional (count and low-level), output voltage (high-level and low-level) and input threshold (low-level and high-level) check test circuit for tables VIII, IX, and X.

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NOTES
1. R1 - R3 = 51Ω ±5%, 1/2 W
2. S1 = SPST switch
3. Adjust item J01 input pulse width, transition times, and pulse amplitude using item D05 with an item J01,PRR of 30 kHz ±50%. Adjust the dc logic levels with item J01 in the single pulse mode using item M02.
4. Connect output loads as specified on figure 10 to pins 5, 6, 13, 14, 18, 19, 24, 25, 26, 31, 33, 34, 38, 39, and 40.
5. For test conditions E, F, J, and K verification of the above logic states is obtained by connecting item M02 to each output pin in succession after input conditions are accomplished.
   L = Low level = 0.0 to 0.4 V.
   H = High level = 2.4 to 4.5 V.
6. Test conditions E1-E3:

   a. Input pulse at point F with point F open.



$t_{TLH} = t_{THL} = 10 \pm 2$ ns

   b. After making all circuit connections for a test condition close switch S1 for a minimum of 100 ns then open switch S1. With item J01 in the single pulse mode, apply 10 pulses to the circuit under test. The output logic states for each input pulse shall be as specified in table XI.

7. Test conditions F1-F9:

   a. The input pulse at point F with point F open shall be as specified in note 6.
   b. After making all circuit connections for a test condition close switch S1 for a minimum of 100 ns then open switch S1. With item J01 in the single pulse mode apply 1 pulse to the circuit under test. The output logic states shall be as specified in table VIII.

8. Test conditions G1-G15 and H1-H12:

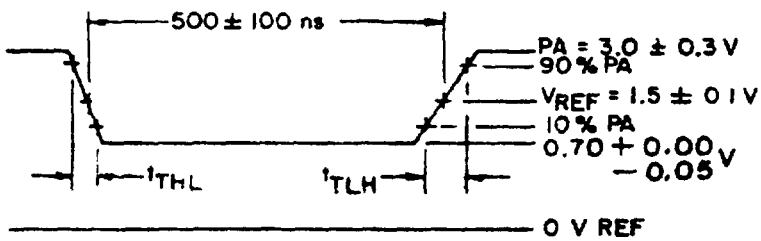
   a. Input pulse at point F with point F open shall be as specified in note 6.
   b. After making all circuit connections for a test condition close switch S1 for a minimum of 100 ns then open switch S1. With item J01 in the single pulse mode apply 1 pulse to the circuit under test. The output voltage level shall be as specified in table IX.

* FIGURE 9. Functional (count and low-level), output voltage (high-level and low-level) and input threshold (low-level and high-level) check test circuit for tables VIII, IX, and X - Continued.

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9. Test conditions J1-J3:

a. Input pulse at point F with point F open.



$t_{TLH} = t_{THL} = 10 \pm 2$ ns

b. Initially switch S1 shall be closed. After making all circuit connections for a test condition open switch S1 for a minimum of 100 ns then close switch S1. With item J01 in the single pulse mode apply 1 pulse to the circuit under test. The output logic states shall be as specified in table X.

10. Test conditions K1-K6

a. The input pulse at point F with point F open shall be as specified in note 6.
b. After making all connections for a test condition close switch S1 for a minimum of 100 ns then open switch S1. With item J01 in the single mode apply 1 pulse to the circuit under test. The output logic states shall be as specified in table X.

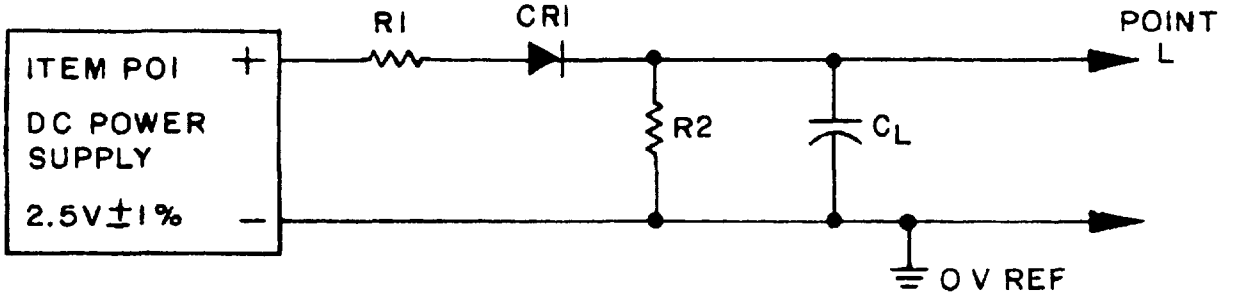
* FIGURE 9. Functional (count and low-level), output voltage (high-level and low-level) and input threshold (low-level and high-level) check test circuit for tables VIII, IX, and X - Continued.

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TABLE XI. Output logic states for test conditions E1-E3, table VIII.

Test number	Total pulses	Q0	Q1	Q2	Q3	TC
Initial state	0	L	L	L	L	L
1	1	H	L	L	L	L
2	2	L	H	L	L	L
3	3	H	H	L	L	L
4	4	L	L	H	L	L
5	5	H	L	H	L	L
6	6	L	H	H	L	L
7	7	H	H	H	L	L
8	8	L	L	L	H	L
9	9	H	L	L	H	H
10	10	L	L	L	L	L

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NOTES
1 CR1 = 1N4150, or equivalent.
2 R1 = Select value for a current flow of 12.8 mA ±1% out of point L with point L held at 0.4 V.
3 R2 = 3.74 kΩ ±1%, 1/8 W.
4 C_L = 100 ±5 pF (including probe and parasitic capacitance)
5 C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions CC and DD.

FIGURE 10 ADV logic load circuit.

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TABLE XII. Input current (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1,2,3 MR Input current, low level	Open	Open	20	Open	-0.55	-5.77	mA	11
L2	1CP "	"	"	21	"	-0.10	-3.46	"	"
L3	1CET "	"	"	7	"	"	"	"	"
L4	1PE "	"	"	27	"	"	"	"	"
L5	2,3 CEP "	"	"	28	"	"	"	"	"
L6	2CET "	"	"	32	"	"	"	"	"
L7	3CET "	"	"	17	"	"	"	"	"
L8	2PE "	"	"	12	"	"	"	"	"
L9	3PE "	"	"	37	"	"	"	"	"
L10	2,3 CP "	"	"	11	"	"	-6.92	"	"
L11	1CEP "	"	"	4	"	"	-1.73	"	"
L12	1PO "	"	"	2	27	"	-1.12	"	"
L13	1P1 "	"	"	22	"	"	"	"	"
L14	1P2 "	"	"	3	"	"	"	"	"
L15	1P3 "	"	"	23	"	"	"	"	"
L16	2PO "	"	"	30	12	"	"	"	"
L17	2P1 "	"	"	29	"	"	"	"	"
L18	2P2 "	"	"	9	"	"	"	"	"
L19	2P3 "	"	"	8	"	"	"	"	"
L20	3P3 "	"	"	35	37	"	"	"	"
L21	3P2 "	"	"	15	37	"	"	"	"

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TABLE XII. Input current (low-level and high-level) test - Continued

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
L22	3P1	Input current, low level	Open	Open	36	37	-0.10	-1.12	mA	11
L23	3P0	Input current, low level	Open	"	16	37	-0.10	-1.12	mA	"
M1	1,2,3 MR	Input current, high level	20	"	Open	2,3,4,7,8,9,11, 12,15,16,17,21, 22,23,27,28,29, 30,32,35,36,37	-188	-324	µA	"
M2	1CP	"	21	"	"	2,3,4,7,20,22, 23,37	0	80	"	"
M3	1CET	"	7	"	"	2,3,4,20,21,22, 23,37	"	"	"	"
M4	1PE	"	27	"	"	2,3,4,7,20,21, 22,23	"	"	"	"
M5	2,3 CEP	"	28	"	"	8,9,11,12,15,16, 17,20,29,30,32, 35,36,37	"	"	"	"
M6	2CET	"	32	"	"	8,9,11,12,20,28, 29,30	"	"	"	"
M7	3CET	"	17	"	"	11,15,16,20,28, 35,36,37	"	"	"	"
M8	2PE	"	12	"	"	8,9,11,20,28,29, 30,32	"	"	"	"
M9	3PE	"	37	"	"	11,15,16,17,20, 28,35,36	"	"	"	"
M10	2,3 CP	"	11	"	"	8,9,12,15,16,17, 20,28,29,30,32, 35,36,37	"	160	"	"
M11	1CEP	"	4	"	"	2,3,7,20,21,22, 23,27	"	40	"	"

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TABLE XII. Input current (low-level and high-level) test - Continued.

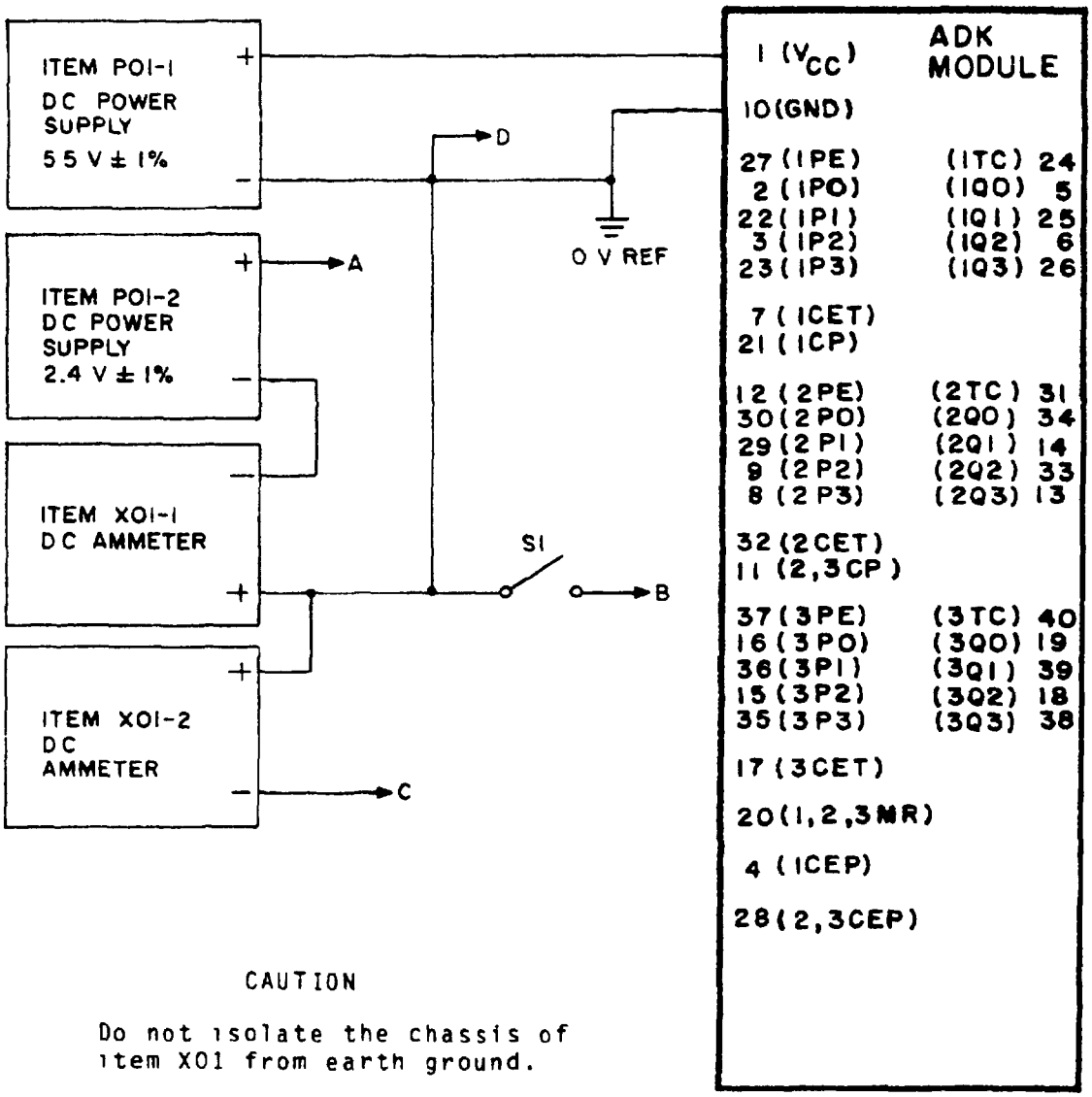
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M12	1PO	Input current, high level	2	Open	Open	3,4,7,20,21,22, 23,27	0	27	µA	11
M13	1P1	"	22	"	"	2,3,4,7,20,21, 23,27	"	"	"	"
M14	1P2	"	3	"	"	2,4,7,20,21,22, 23,27	"	"	"	"
M15	1P3	"	23	"	"	2,3,4,7,20,21, 22,27	"	"	"	"
M16	2PO	"	30	"	"	8,9,11,12,20, 28,29,32,37	"	"	"	"
M17	2P1	"	29	"	"	8,9,11,12,20, 28,30,32,37	"	"	"	"
M18	2P2	"	9	"	"	8,11,12,20,28, 29,30,32,37	"	"	"	"
M19	2P3	"	8	"	"	9,11,12,20,28, 29,30,32,37	"	"	"	"
M20	3P3	"	35	"	"	11,15,16,17,20, 28,36,37	"	"	"	"
M21	3P2	"	15	"	"	11,16,17,20,28, 35,36,37	"	"	"	"
M22	3P1	"	36	"	"	11,15,16,17,20, 28,35,37	"	"	"	"
M23	3PO	"	16	"	"	11,15,17,20,28, 35,36,37	"	"	"	"

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TABLE XIII. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Q3 Output short circuit current	Open	11,21	26	27	-20	-80	mA	11
N2	1Q2 "	"	"	6	"	"	"	"	"
N3	1Q1 "	"	"	25	"	"	"	"	"
N4	1Q0 "	"	"	5	"	"	"	"	"
N5	1TC "	"	"	24	3,22,27	"	"	"	"
N6	2Q3 "	"	"	13	12	"	"	"	"
N7	2Q2 "	"	"	33	"	"	"	"	"
N8	2Q1 "	"	"	14	"	"	"	"	"
N9	2Q0 "	"	"	34	"	"	"	"	"
N10	2TC "	"	"	31	9,12,29	"	"	"	"
N11	3Q2 "	"	"	18	37	"	"	"	"
N12	3Q3 "	"	"	38	"	"	"	"	"
N13	3Q1 "	"	"	39	"	"	"	"	"
N14	3Q0 "	"	"	19	"	"	"	"	"
N15	3TC "	"	"	40	15,36,37	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. S1 = SPST switch.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
3. After making all circuit connections for a test condition except point C, close switch S1 for a minimum of 100 ns then open switch S1. Then connect point C and take measurement.
4. Only one output shall be shorted at a time and the measurement taken within 10 seconds.

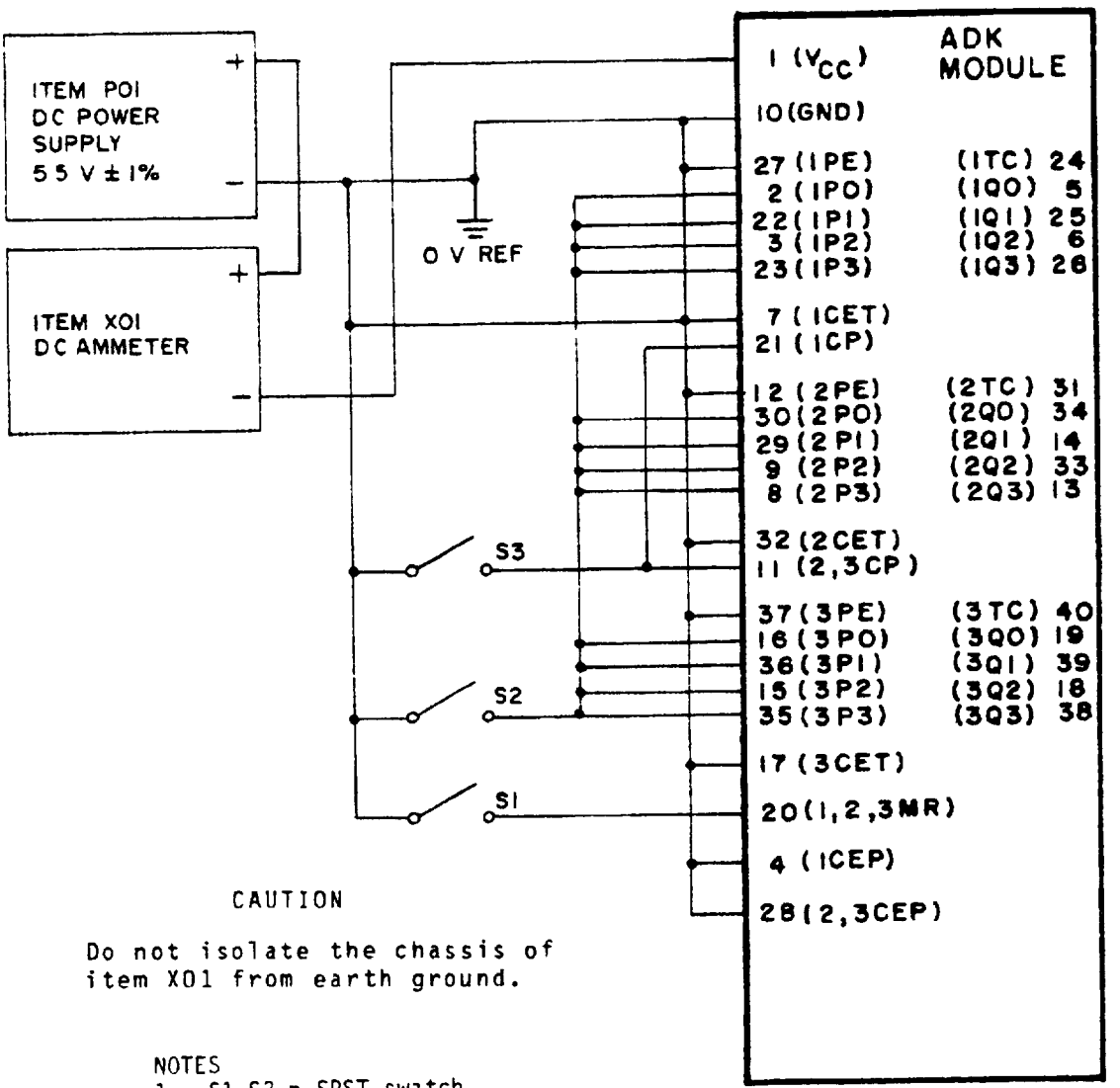
FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for tables XII and XIII.

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*TABLE XIV. Power supply current (high-level and low-level) test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P1	Power supply current, high level	10	300	mA	12
P2	Power supply current, low level	10	300	mA	12

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

NOTES
1. S1-S3 = SPST switch.
2. Power supply current - all outputs high.

   a. Switch S1 open.
   b. Initially switch S2 shall be open and switch S3 closed.
   c Open switch S3 for 100 ns minimum then close switches S3 and S2
   d. Take measurement.

3 Power supply current - all outputs low

   a. Close switches S1, S2, and S3.
   b. Take measurement.

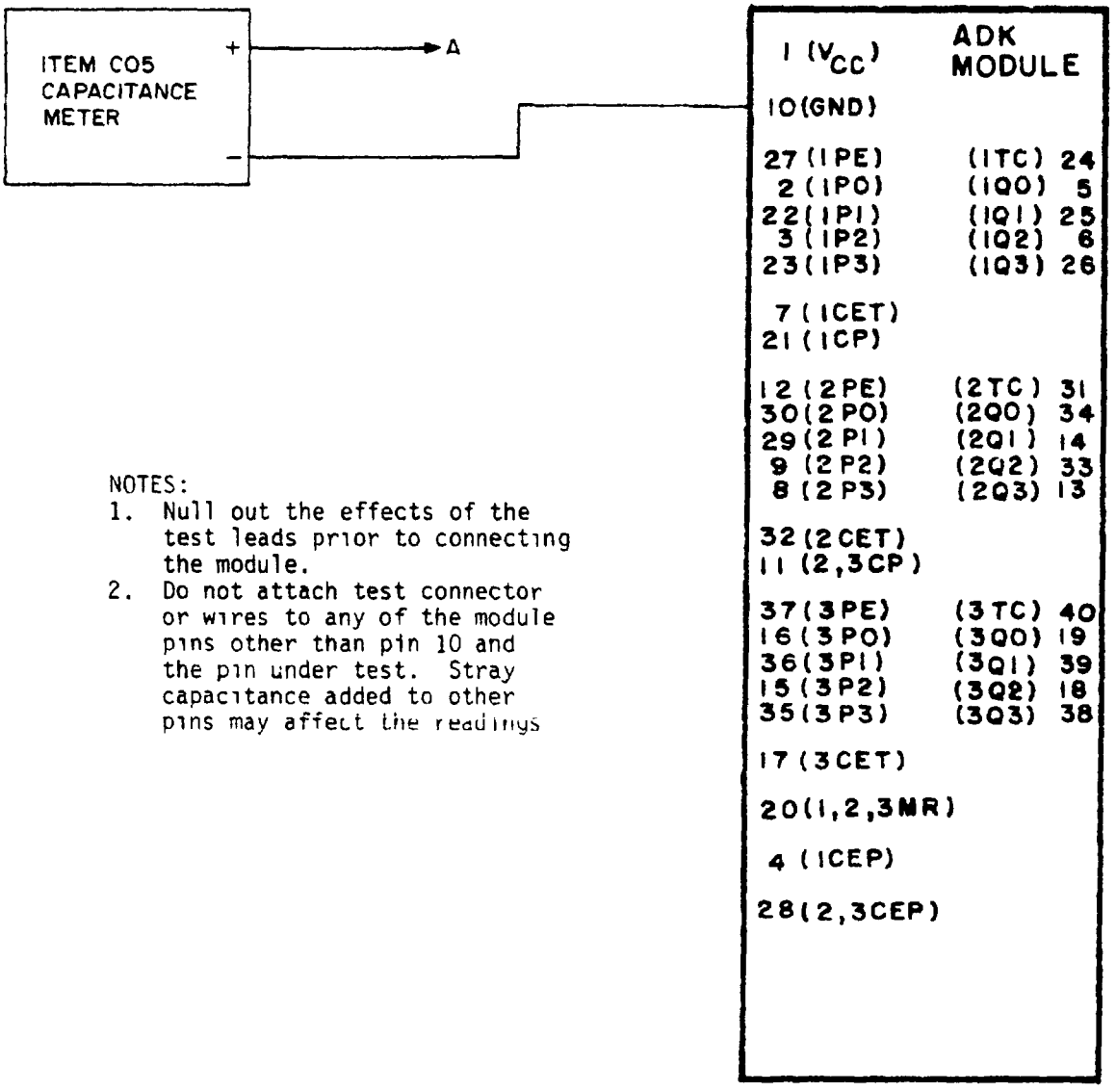
* FIGURE 12. Power supply current (high-level and low-level) test circuit for table XIV.

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TABLE XV Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	1,2,3 MR Input capacitance	20	3	45	pF	13
Q2	2,3 CEP "	28	2	30	"	"
Q3	2,3 CP "	11	"	30	"	"
Q4	1PE "	27	"	15	"	"
Q5	1CP "	21	"	"	"	"
Q6	1P0 "	2	"	"	"	"
Q7	1P1 "	22	"	"	"	"
Q8	1P2 "	3	"	"	"	"
Q9	1P3 "	23	"	"	"	"
Q10	1CET "	7	"	"	"	"
Q11	1CEP "	4	"	"	"	"
Q12	2PE "	12	"	"	"	"
Q13	2P0 "	30	"	"	"	"
Q14	2P1 "	29	"	"	"	"
Q15	2P2 "	9	"	"	"	"
Q16	2P3 "	8	"	"	"	"
Q17	2CET "	32	"	"	"	"
Q18	3P0 "	16	"	"	"	"
Q19	3P1 "	36	"	"	"	"
Q20	3P2 "	15	"	"	"	"
Q21	3P3 "	35	"	"	"	"
Q22	3CET "	17	"	"	"	"
Q23	3PE "	37	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings

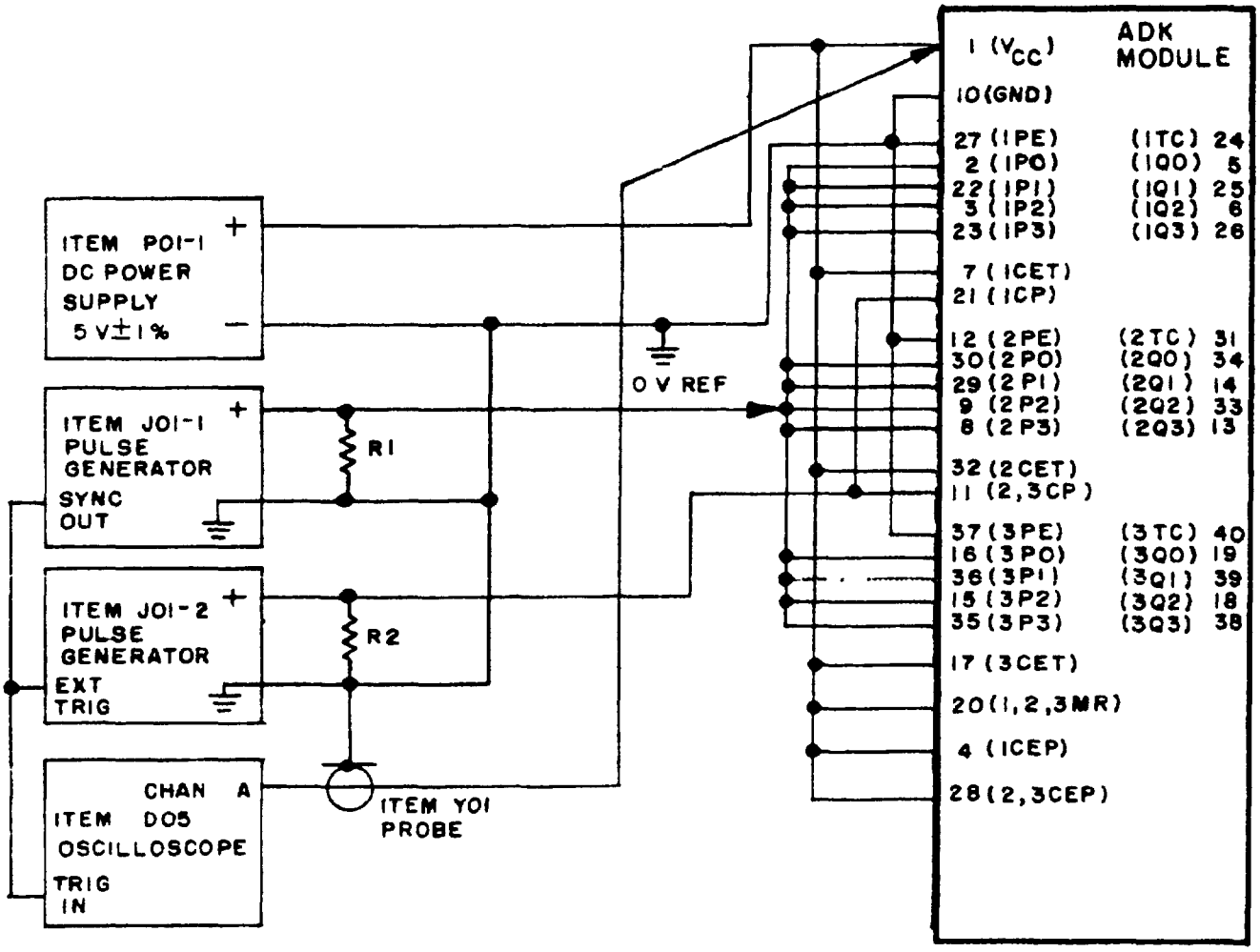
FIGURE 13. Input capacitance test circuit for table XV.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	14,15

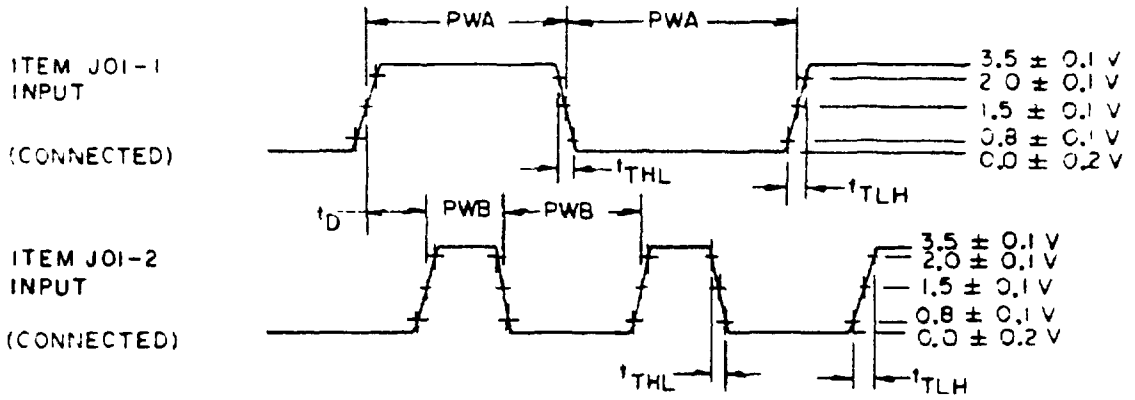
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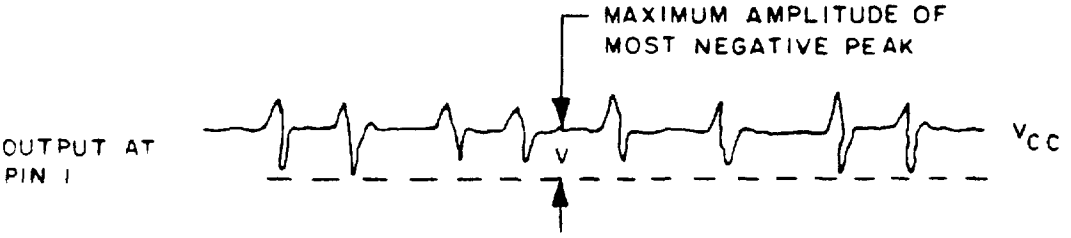


NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as specified on figure 10.
3. Peak noise on V_{CC} shall be less than 400 mV.
4. Item J01-2 is operated in the double pulse mode.

FIGURE 14. Power supply transient amplitude test circuit for table XVI.

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NOTES.
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 250 ±25 ns.
   t_D = 125 ±25 ns
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
2. The value V is measured from the V_{CC} baseline to the most negative excursion of the waveform

FIGURE 15 Power supply transient amplitude test circuit waveforms for table XVI.

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TABLE XVII. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test.

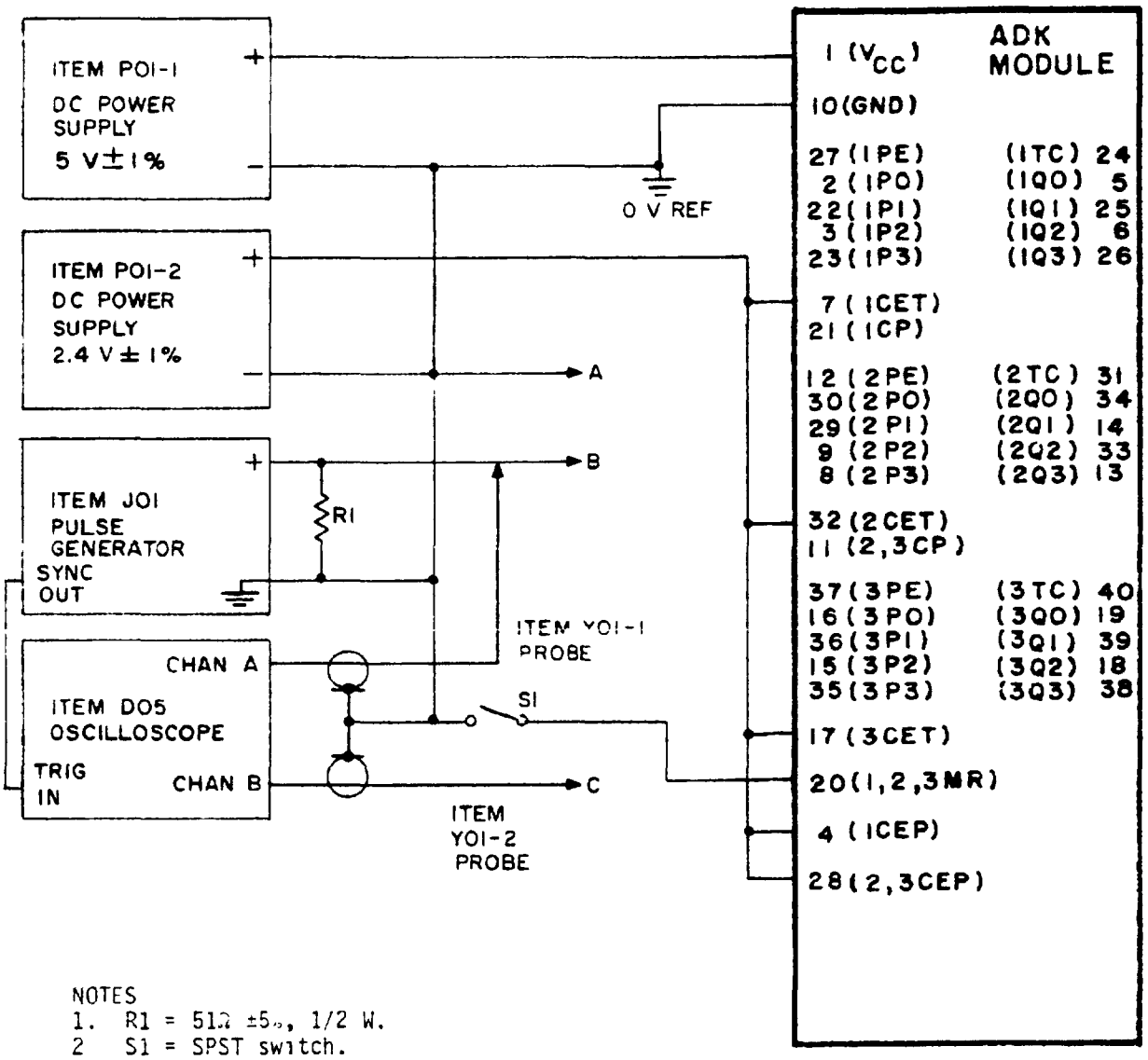
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient				Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Transition time		Propagation delay time			
						Min	Max	Min	Max		
S1,U1	1CP to 1TC Transition time (t_{THL}) and propagation delay time (t_{PHL})	3,22	21	24	26 to 27	2	20	2	45	ns	16,17
S2,U2	2,3 CP to 2TC "	9,29	11	31	13 to 12	"	"	"	"	"	"
S3,U3	2,3 CP to 3TC "	15,36	11	40	38 to 37	"	"	"	"	"	"
S4,U4	1CP to 1Q0 "	3,22	21	5	26 to 27	"	"	"	40	"	"
S5,U5	1CP to 1Q3 "	3,22	"	26	5 to 27	"	"	"	"	"	"
S6,U6	1CP to 1Q1 "	23	"	25	6 to 27	"	"	"	"	"	"
S7,U7	1CP to 1Q2 "	23	"	6	25 to 27	"	"	"	"	"	"
S8,U8	2,3 CP to 2Q0 "	9,29	11	34	13 to 12	"	"	"	"	"	"
S9,U9	2,3 CP to 2Q3 "	9,29	"	13	34 to 12	"	"	"	"	"	"
S10,U10	2,3 CP to 2Q1 "	8	"	14	33 to 12	"	"	"	"	"	"
S11,U11	2,3 CP to 2Q2 "	8	"	33	14 to 12	"	"	"	"	"	"
S12,U12	2,3 CP to 3Q0 "	15,36	"	19	38 to 37	"	"	"	"	"	"
S13,U13	2,3 CP to 3Q3 "	15,36	"	38	19 to 37	"	"	"	"	"	"
S14,U14	2,3 CP to 3Q1 "	35	"	39	18 to 37	"	"	"	"	"	"
S15,U15	2,3 CP to 3Q2 "	35	"	18	39 to 37	"	"	"	"	"	"
T1,V1	1CP to 1TC Transition time (t_{TLH}) and propagation delay time (t_{PLH})	2,3,22	21	24	26 to 27	5	35	4	65	"	"
T2,V2	2,3 CP to 2TC "	9,29,30	11	31	13 to 12	"	"	"	"	"	"
T3,V3	2,3 CP to 3TC "	15,16,36	11	40	38 to 37	"	"	"	"	"	"
T4,V4	1CP to 1Q0 "	2,3,22	21	5	26 to 27	"	"	"	45	"	"
T5,V5	1CP to 1Q3 "	23	"	26	5 to 27	"	"	"	"	"	"
T6,V6	1CP to 1Q1 "	3,22,23	"	25	5 to 27	"	"	"	"	"	"

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TABLE XVII Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient				Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Transition time		Propagation delay time			
						Min	Max	Min	Max		
T7,V7	1CP to 1Q2 Transition time (t_{TLH}) and propagation delay time (t_{PLH})	3,23	21	5	25 to 27	5	35	4	45	ns	16,17
T8,V8	2,3 CP to 2Q0 "	9,29,30	11	31	13 to 12	"	"	"	"	"	"
T9,V9	2,3 CP to 2Q3 "	8	"	13	34 to 12	"	"	"	"	"	"
T10,V10	2,3 CP to 2Q1 "	8,9,29	"	14	34 to 12	"	"	"	"	"	"
T11,V11	2,3 CP to 2Q2 "	8,9	"	33	14 to 12	"	"	"	"	"	"
T12,V12	2,3 CP to 3Q0 "	15,16,36	"	19	38 to 37	"	"	"	"	"	"
T13,V13	2,3 CP to 3Q3 "	35	"	38	19 to 37	"	"	"	"	"	"
T14,V14	2,3 CP to 3Q1 "	15,35,36	"	39	19 to 37	"	"	"	"	"	"
T15,V15	2,3 CP to 3Q2 "	15,35	"	18	39 to 37	"	"	"	"	"	"

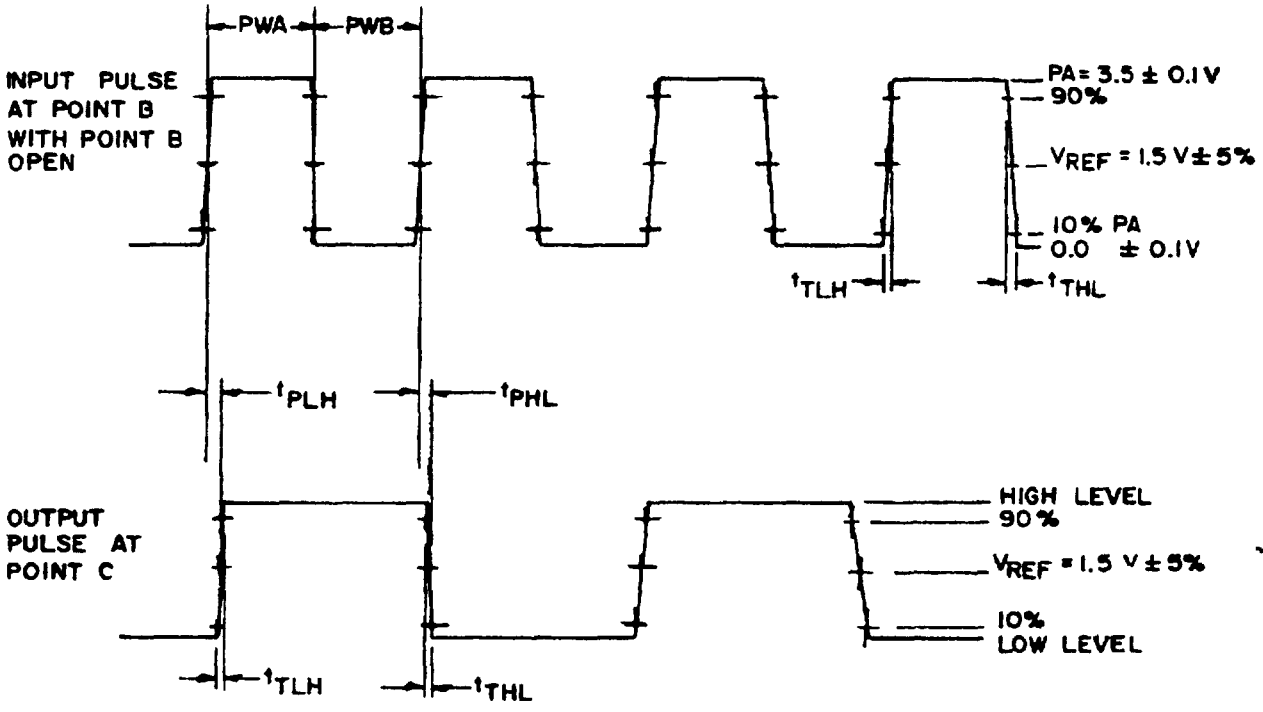
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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. Connect output loads as specified on figure 10 to pins 5, 6, 13, 14, 18, 19, 24, 25, 26, 31, 33, 34, 38, 39, and 40.
4. After making all circuit connections for a test condition, momentarily close switch S1 for a minimum of 100 ns

FIGURE 16. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XVII.

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NOTES.
1. Input pulse parameters.
   PWA = 500 ±50 ns.
   PWB = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Before making the measurements specified in table XVII, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.

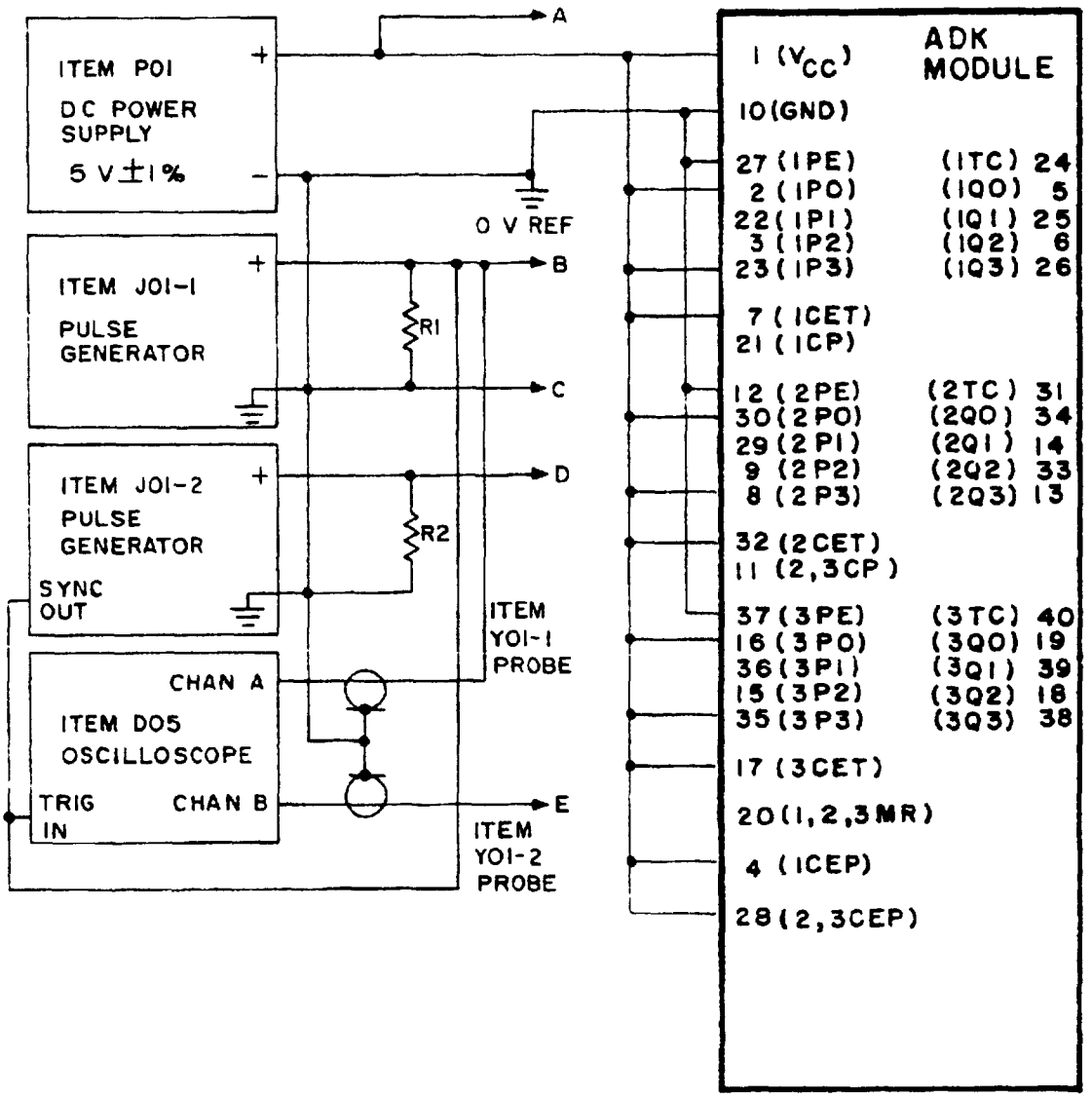
FIGURE 17. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XVII.

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TABLE XVIII. Propagation delay time (t_{PHL}) and minimum MR pulse width and recovery time check test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	MR to 1Q3 Propagation delay time (t_{PHL}) and minimum MR pulse width and recovery time check	3,22	20	Open	21	26	5	65	ns	18,19
W2	MR to 2Q3 "	9,29	"	"	11	13	"	"	"	"
W3	MR to 3Q3 "	15,36	"	"	11	38	"	"	"	"
W4	MR to 1TC "	Open	"	22,3	21	24	"	"	"	"
W5	MR to 1Q0 "	3,22	"	Open	"	5	"	"	"	"
W6	MR to 1Q1 "	"	"	"	"	25	"	"	"	"
W7	MR to 1Q2 "	"	"	"	"	6	"	"	"	"
W8	MR to 2TC "	Open	"	9,29	11	31	"	"	"	"
W9	MR to 2Q0 "	9,29	"	Open	"	34	"	"	"	"
W10	MR to 2Q1 "	"	"	"	"	14	"	"	"	"
W11	MR to 2Q2 "	"	"	"	"	33	"	"	"	"
W12	MR to 3TC "	Open	"	15,36	"	40	"	"	"	"
W13	MR to 3Q0 "	15,36	"	Open	"	19	"	"	"	"
W14	MR to 3Q1 "	"	"	"	"	39	"	"	"	"
W15	MR to 3Q2 "	"	"	"	"	18	"	"	"	"

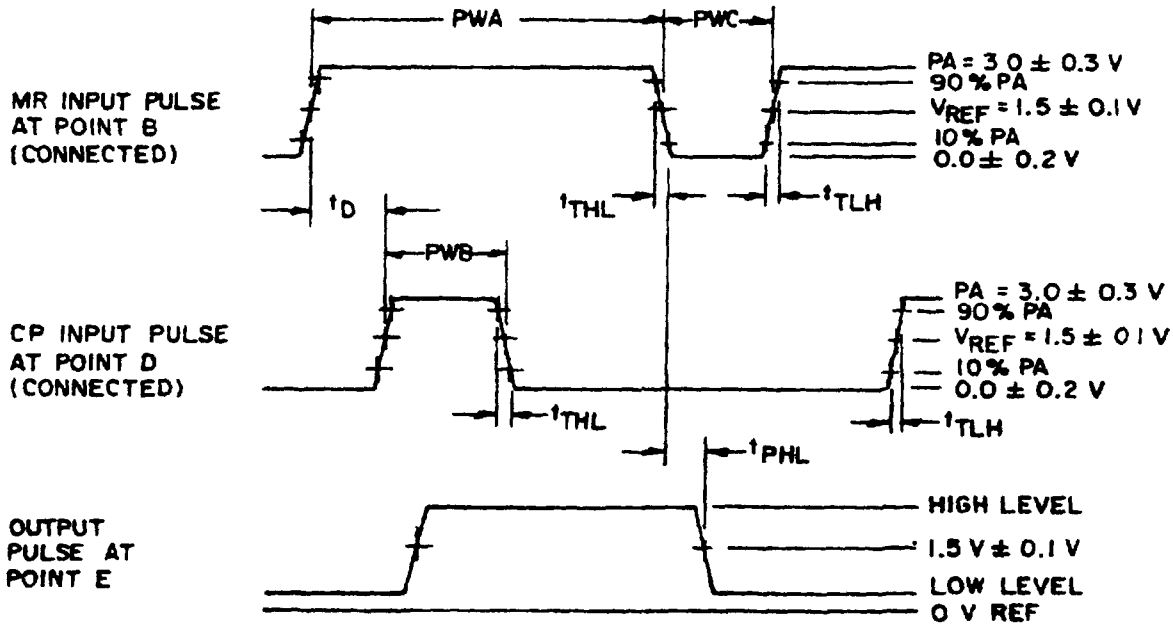
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Connect output loads as specified on figure 10 to pins 5, 6, 13, 14, 18, 19, 24, 25, 26, 31, 33, 34, 38, 39, and 40.

FIGURE 18. Propagation delay time (t_{PHL}) and minimum MR pulse width and recovery time check test circuit for table XVIII.

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NOTES.
1. Input pulse parameters.
   PWA = 300 ±30 ns.
   PWB = 100 ±10 ns.
   PWC = 35 $^{+3}_{-0}$ ns (minimum MR pulse width).
   t_D = 20 $^{+2}_{-0}$ ns (minimum MR recovery time).
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to ground.

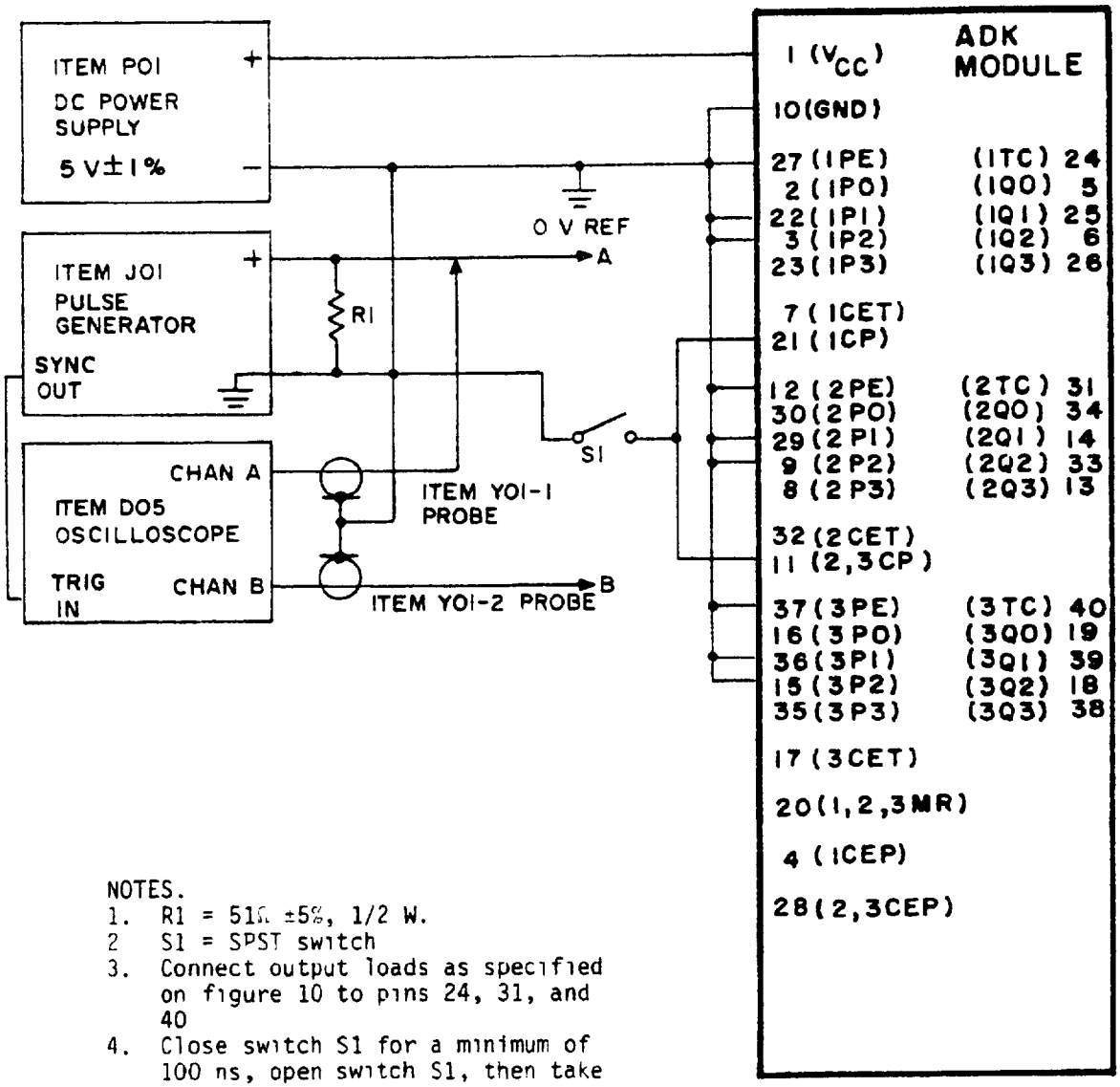
FIGURE 19. Propagation delay time (t_{PHL}) and minimum MR pulse width and recovery time check test circuit waveforms for table XVIII.

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TABLE XIX. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	1CET to 1TC Propagation delay time (t_{PHL})	7	24	2	45	ns	20,21
X2	2CET to 2TC "	32	31	"	"	"	"
X3	3CET to 3TC "	17	40	"	"	"	"
Y1	1CET to 1TC Propagation delay time (t_{PLH})	7	24	"	"	"	"
Y2	2CET to 2TC "	32	31	"	"	"	"
Y3	3CET to 3TC "	17	40	"	"	"	"

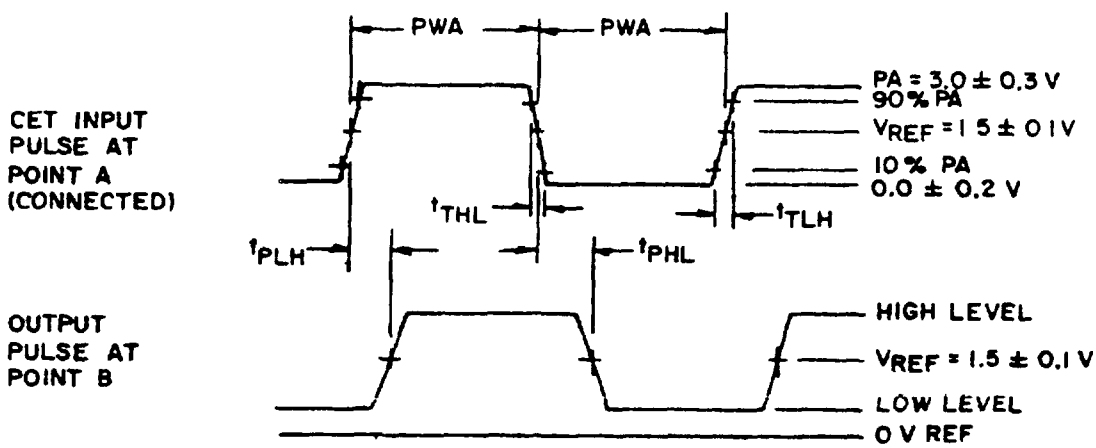
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:46Z
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NOTES.
1. R1 = 51Ω ±5%, 1/2 W.
2 S1 = SPST switch
3. Connect output loads as specified on figure 10 to pins 24, 31, and 40
4. Close switch S1 for a minimum of 100 ns, open switch S1, then take measurement.

FIGURE 20. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XIX.

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NOTES
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. All voltages are referenced to ground.

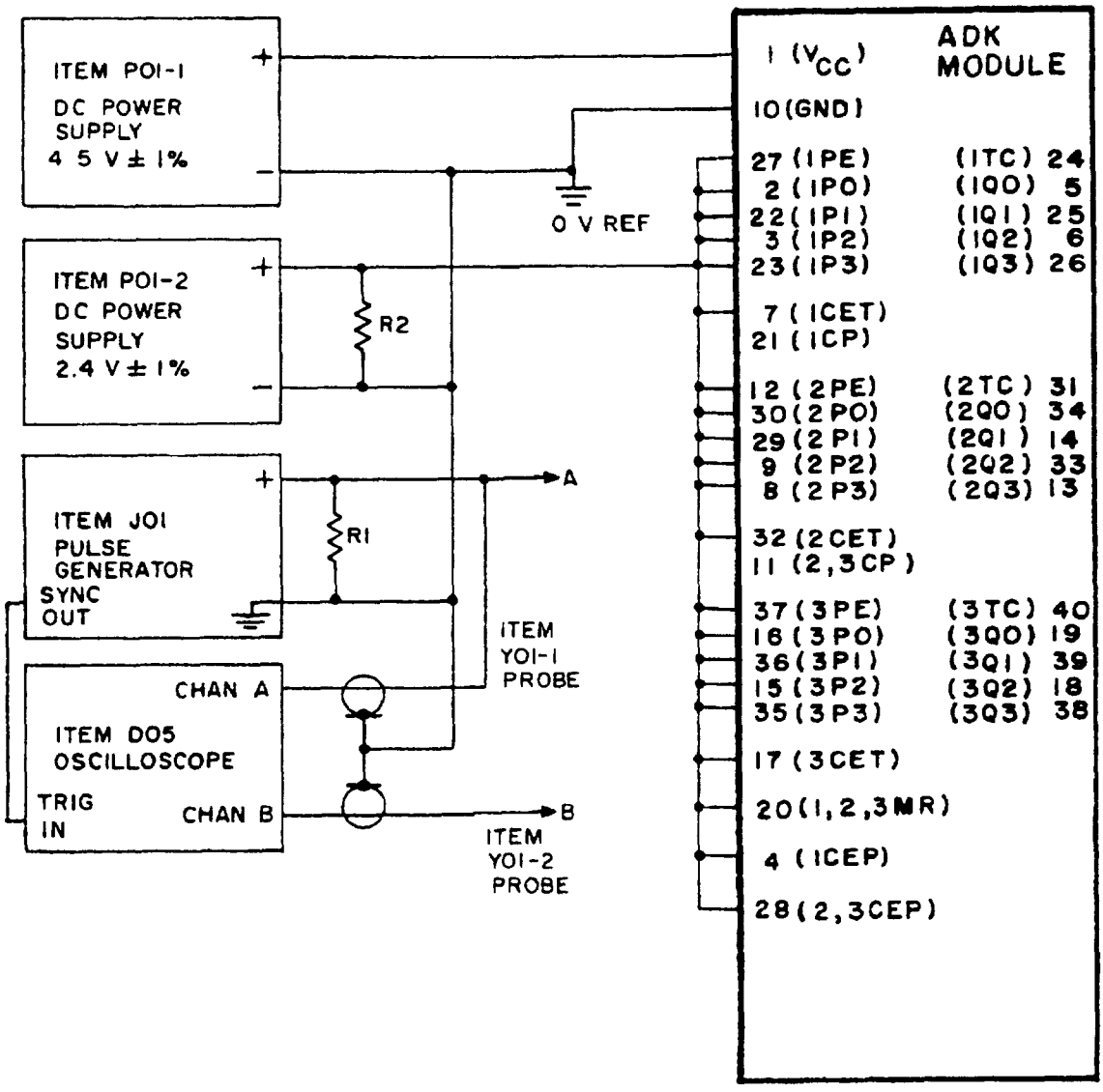
FIGURE 21. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XIX.

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TABLE XX Maximum counting frequency (minimum clock pulse width) check test

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z1	Maximum counting frequency (minimum clock pulse width) check (PW # 1)	21	26	85	115	ns	22,23
Z2	"	11	13	"	"	"	"
Z3	"	11	38	"	"	"	"
Z4	Maximum counting frequency (minimum clock pulse width) check (PW # 2)	21	26	385	415	"	"
Z5	"	11	13	"	"	"	"
Z6	"	11	38	"	"	"	"

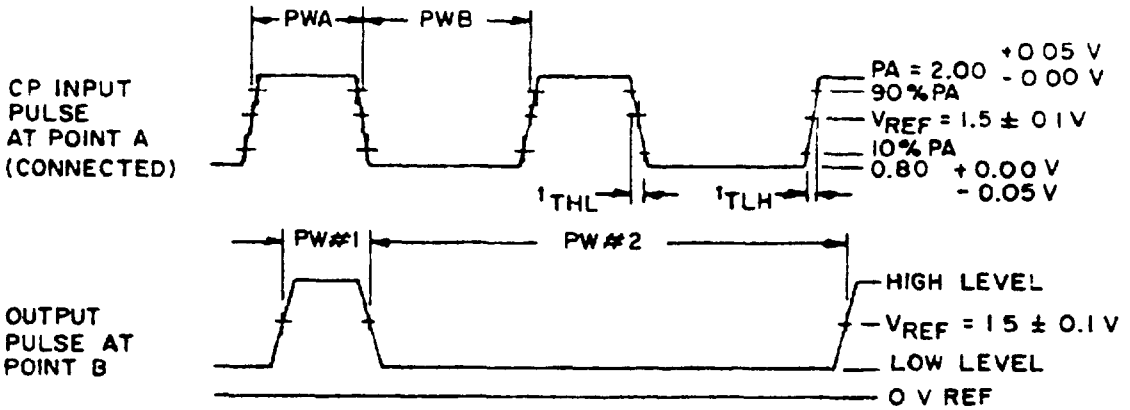
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:46Z
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NOTES:
1. R1 = R2 = 51.) ±5%, 1/2 W.
2. Connect output loads as specified on figure 10 to pins 5, 6, 13, 14, 18, 19, 24, 25, 26, 31, 33, 34, 38, 39, and 40.

FIGURE 22. Maximum counting frequency (minimum clock pulse width) check test circuit for table XX.

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NOTES
1. Input pulse parameters
   PWA = 20 +2 -0 ns
   PWB = 30 +3 -0 ns
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns
2. All voltages are referenced to ground.
3. This test verifies a minimum clock frequency of [illegible] under threshold clock conditions.

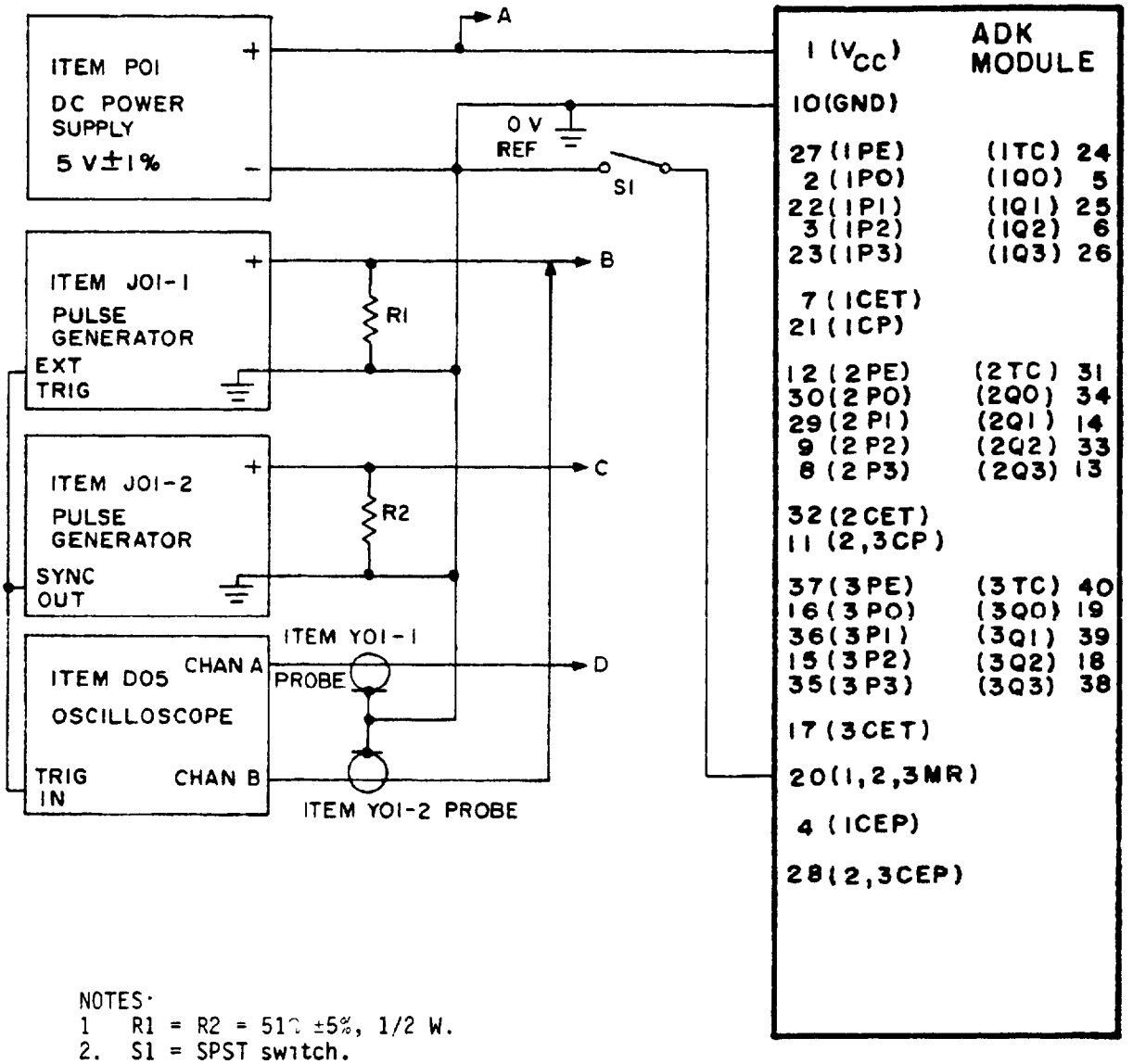
FIGURE 23. Maximum counting frequency (minimum clock pulse width) check test circuit waveforms for table XX.

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*TABLE XXI Setup and hold time (CET and CEP) (measuring output PRP) check test

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
AA1	1CET	Setup and hold time (CET and CEP) (measuring output PRP) check	4	7	21	26	1 9	2.1	µs	24,25
AA2	2CET	"	28	32	11	13	"	"	"	"
AA3	3CET	"	28	17	11	38	"	"	"	"
AA4	1CEP	"	7	4	21	26	"	"	"	"
AA5	2,3 CEP	"	32	28	11	13	"	"	"	"
AA6	2,3 CEP	"	17	28	11	38	"	"	"	"

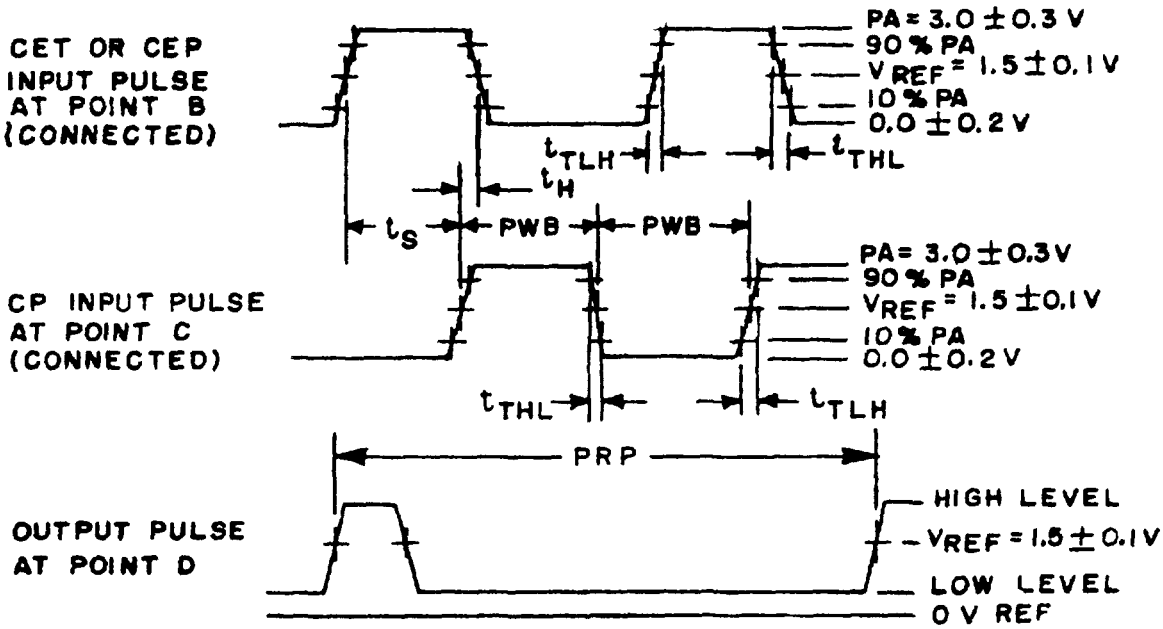
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. Connect output loads as specified on figure 10 to pins 13, 26, and 38.
4. After all circuit connections are made, close switch S1 for a minimum of 100 ns, then open it and make measurement.

* FIGURE 24. Setup and hold time (CET and CEP) (measuring output period) check test circuit for table XXI.

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NOTES:
1. Input pulse parameters.
   $t_S = 30 \, ^{+2}_{-0}$ ns .
   $t_H = 0 \, ^{+2}_{-0}$ ns
   PWB = 100 ±5 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. The output waveform time scale is not the same as the two input waveforms.
3. All voltages are referenced to ground.
4. PRP shall be as specified in table XXI.

FIGURE 25. Setup and hold time (CET and CEP) (measuring output period) check test circuit waveforms for table XXI.

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*TABLE XXII. Setup and hold time (P and PE) (measuring t_{PLH} and t_{PHL}) check test.

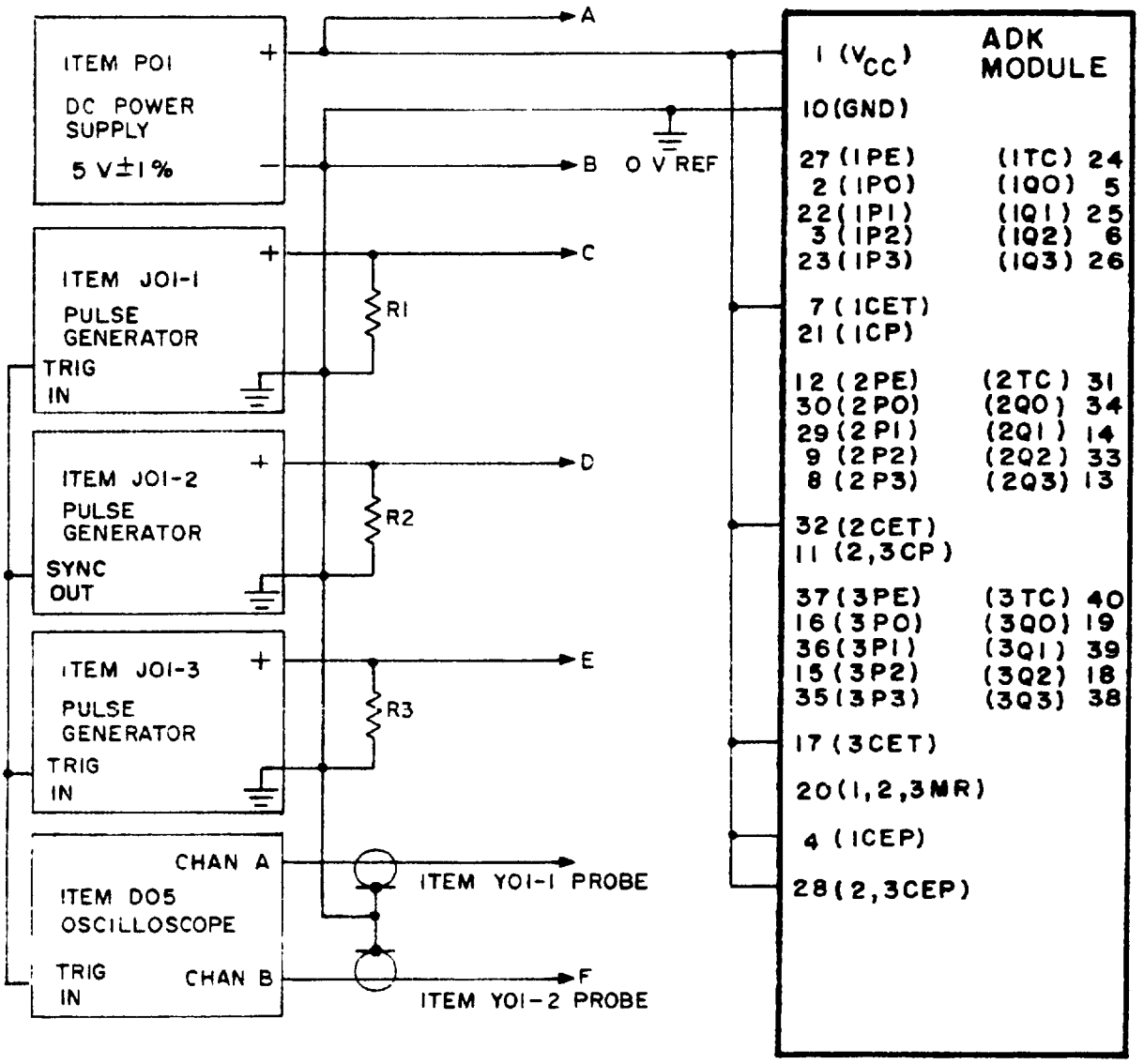
Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
BB1	1PO Setup and hold time (P) (measuring t_{PLH}), high level, check	Open	27	2	21	20	5	5	45	ns	26,27
BB2	1P1 "	"	"	22	"	"	25	"	"	"	"
BB3	1P2 "	"	"	3	"	"	6	"	"	"	"
BB4	1P3 "	"	"	23	"	"	26	"	"	"	"
BB5	2PO "	"	12	30	11	"	34	"	"	"	"
BB6	2P1 "	"	"	29	"	"	14	"	"	"	"
BB7	2P2 "	"	"	9	"	"	33	"	"	"	"
BB8	2P3 "	"	"	8	"	"	13	"	"	"	"
BB9	3PO "	"	37	16	"	"	19	"	"	"	"
BB10	3P1 "	"	"	36	"	"	39	"	"	"	"
BB11	3P2 "	"	"	15	"	"	18	"	"	"	"
BB12	3P3 "	"	"	35	"	"	38	"	"	"	"
BB13	1PE Setup and hold time (PE) (measuring t_{PLH}), low level, check	23	Open	27	21	"	26	"	"	"	26,28
BB14	2PE "	8	"	12	11	"	13	"	"	"	"
BB15	3PE "	35	"	37	11	"	38	"	"	"	"
BB16	1PO Setup and hold time (P) (measuring t_{PHL}) low level, check	Open	"	27	2	21	5	"	"	"	26,29
BB17	1P1 "	"	"	"	22	"	25	"	"	"	"
BB18	1P2 "	"	"	"	3	"	6	"	"	"	"
BB19	1P3 "	"	"	"	23	"	26	"	"	"	"
BB20	2PO "	"	"	12	30	11	34	"	"	"	"

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*TABLE XXII. Setup and hold time (P and PE) (measuring t_{PLH} and t_{PHL}) check test - Continued.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
BB21	2P1 Setup and hold time (P) (measuring t_{PHL}), low level, check	Open	Open	1/	29	11	14	5	45	ns	26,29
BB22	2P2 "	"	"		9	"	33	"	"	"	"
BB23	2P3 "	"	"	"	8	"	13	"	"	"	"
BB24	3P0 "	"	"	37	16	"	19	"	"	"	"
BB25	3P1 "	"	"	"	36	"	39	"	"	"	"
BB26	3P2 "	"	"	"	15	"	18	"	"	"	"
BB27	3P3 "	"	"	"	35	"	38	"	"	"	"

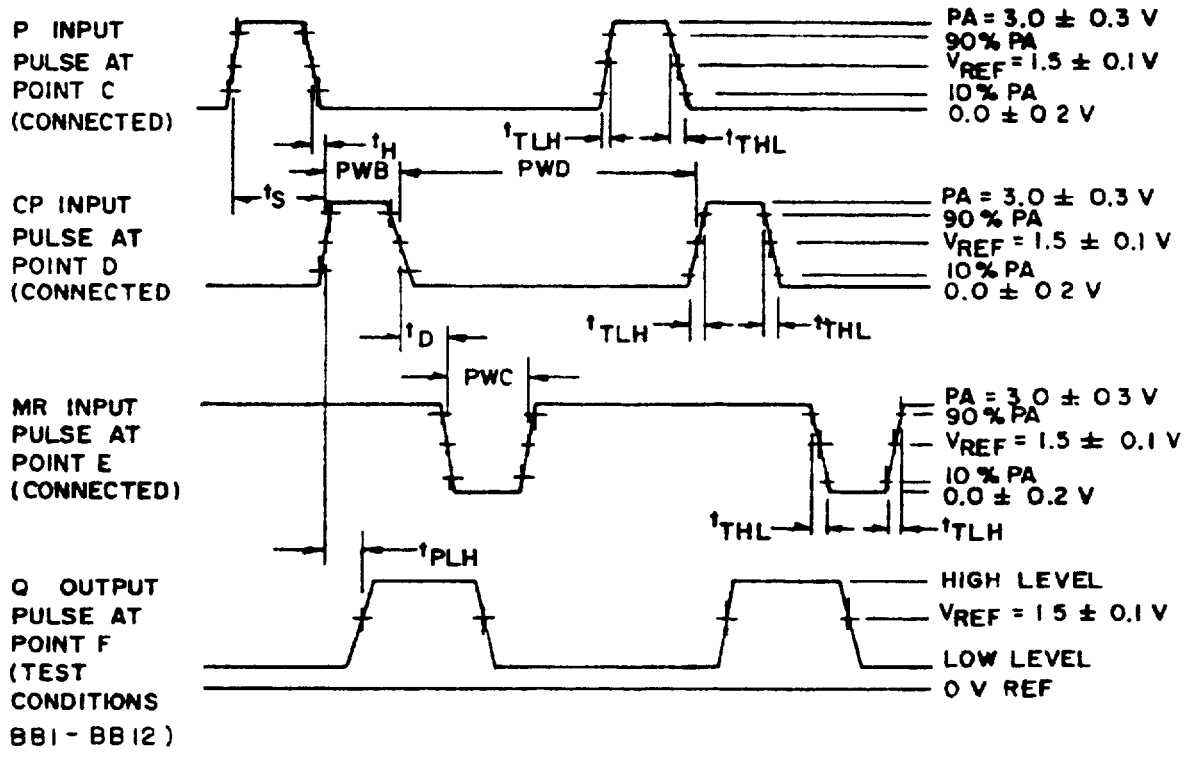
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs as specified on figure 10.
3. For test conditions BB16-BB27 only, place items J01-1 and J01-3 in the double pulse mode to obtain specified waveforms

* FIGURE 26. Setup and hold time (P and PE) (measuring t_{PLH} and t_{PHL}) check test circuit for table XXII.

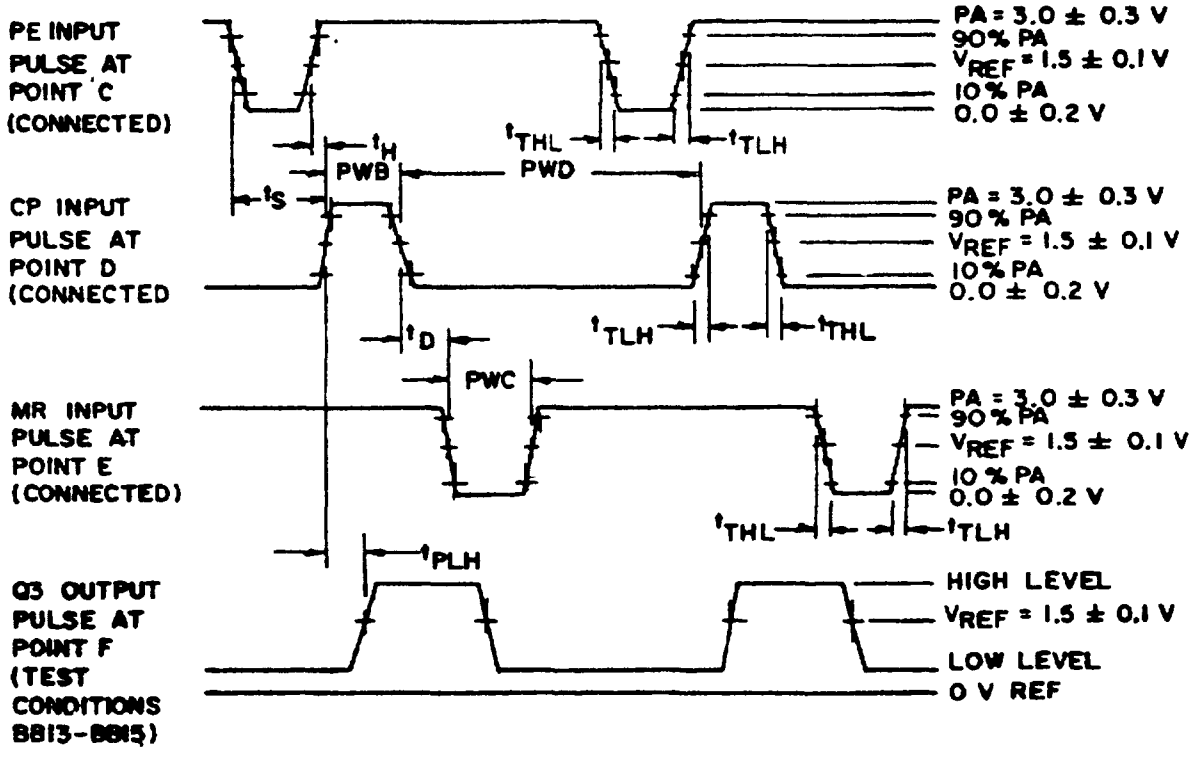
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NOTES
1. Input pulse parameters:
   PWB = 300 ±30 ns.
   PWC = 100 ±10 ns
   PWD = 600 ±30 ns
   $t_S = 38 \,^{+2}_{-0}$ ns .
   $t_H = 0 \,^{+2}_{-0}$ ns .
   $t_D = 100 \pm 10$ ns
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. All voltages are referenced to ground.
3. t_{PLH} shall be as specified in table XXII.

* FIGURE 27. Setup and hold time (P) (measuring t_{PLH}) check test circuit waveforms for table XXII.

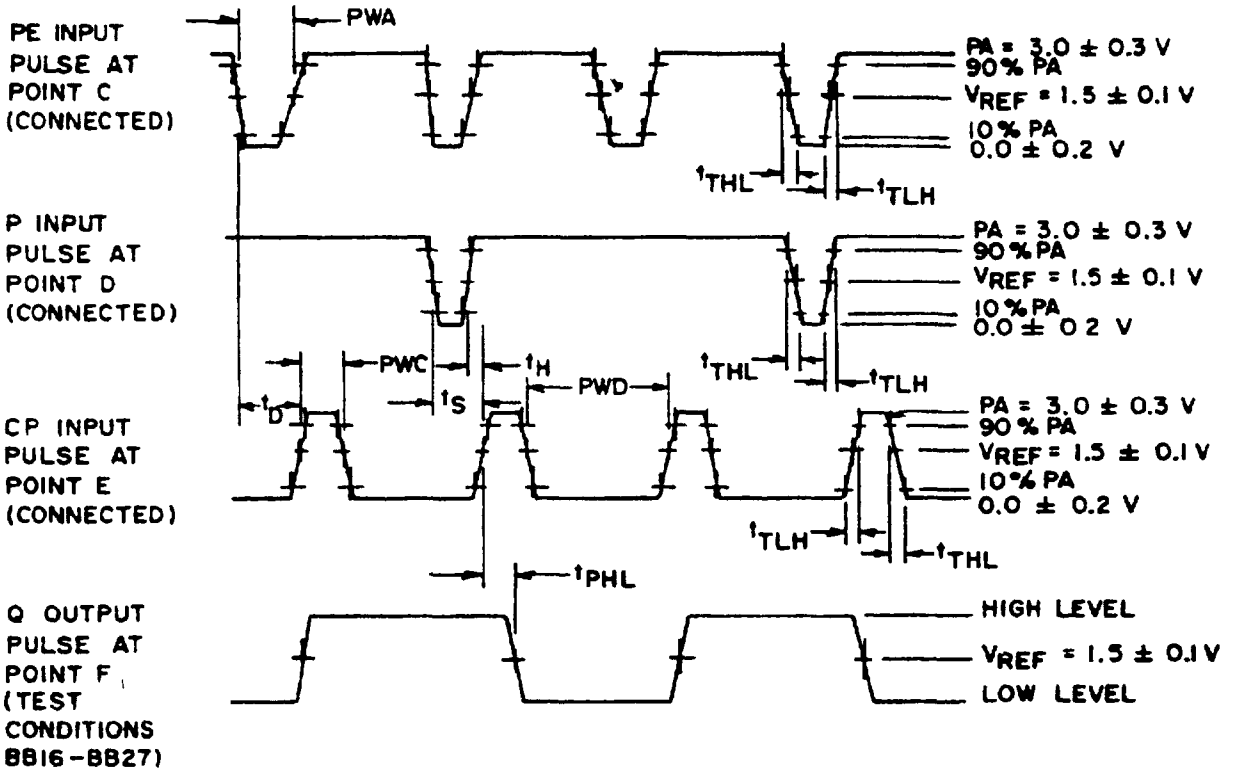
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NOTES
1. Input pulse parameters
   PWB = 300 ±30 ns.
   PWC = 100 ±10 ns.
   PWD = 600 ±30 ns.
   $t_S = 45 \begin{smallmatrix}+2\\-0\end{smallmatrix}$ ns
   $t_H = 0 \begin{smallmatrix}+2\\-0\end{smallmatrix}$ ns .
   $t_D = 100 \pm 10$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. All voltages are referenced to ground.
3. t_{PLH} shall be as specified in table XXII.

* FIGURE 28. Setup and hold time (PE) (measuring t_{PLH}) check test circuit waveforms for table XXII.

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NOTES.
1. Input pulse parameters
   PWA = 150 ±10 ns
   PWC = 300 ±30 ns.
   PWD = 300 ±30 ns.
   t_S = 38 ±1 ns
   t_H = 0 ±1 ns.
   t_D = 125 ±10 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to ground.
3. t_{PHL} shall be as specified in table XXII.

* FIGURE 29. Setup and hold time (P) (measuring t_{PHL}) check test circuit waveforms for table XXII.

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TABLE XXIII. Output crosstalk (high-level, low-level, and duration) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	D to pin	E to pin	Min	Max		
CC1	1Q0 Output crosstalk, high level	2	3,8,9,15,16,22, 23,29,30,35,36	5	Open	0	400	mV	30,31
CC2	1Q1 "	22	2,3,8,9,15,16, 23,29,30,35,36	25	"	"	"	"	"
CC3	1Q2 "	3	2,8,9,15,16,22, 23,29,30,35,36	6	"	"	"	"	"
CC4	1Q3 "	23	2,3,8,9,15,16, 22,29,30,35,36	26	"	"	"	"	"
CC5	1TC "	2,23	8,9,15,16,29, 30,35,36	24	3,22	"	"	"	"
CC6	2Q0 "	30	2,3,8,9,15,16, 22,23,29,35,36	34	Open	"	"	"	"
CC7	2Q1 "	29	2,3,8,9,15,16, 22,23,30,35,36	14	"	"	"	"	"
CC8	2Q2 "	9	2,3,8,15,16,22, 23,29,30,35,36	33	"	"	"	"	"
CC9	2Q3 "	8	2,3,9,15,16,22, 23,29,30,35,36	13	"	"	"	"	"
CC10	2TC "	8,30	2,3,15,16,22, 23,35,36	31	9,29	"	"	"	"
CC11	3Q0 "	16	2,3,8,9,15,22, 23,29,30,35,36	19	Open	"	"	"	"
CC12	3Q1 "	36	2,3,8,9,15,16, 22,23,29,30,35	39	"	"	"	"	"
CC13	3Q2 "	15	2,3,8,9,16,22, 23,29,30,35,36	18	"	"	"	"	"
CC14	3Q3 "	35	2,3,8,9,15,16, 22,23,29,30,36	38	"	"	"	"	"
CC15	3TC "	16,35	2,3,8,9,22,23, 29,30,	40	15,36	"	"	"	"

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TABLE XXIII. Output crosstalk (high-level, low-level, and duration) test - Continued.

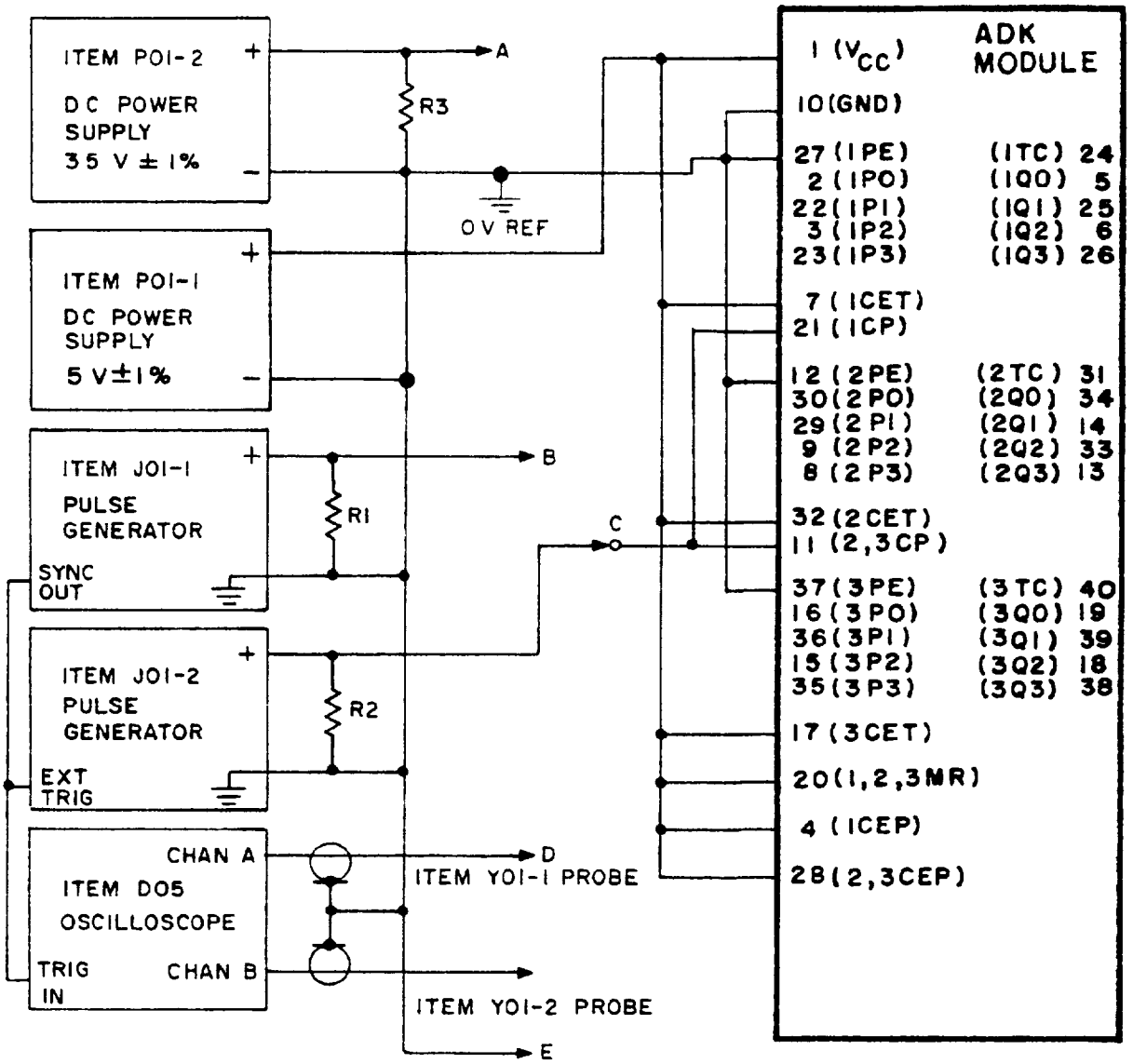
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	D to pin	E to pin	Min	Max		
DD1	1Q0 Output crosstalk, low level	Open	3,8,9,15,16,22, 23,29,30,35,36	5	2	0	800	mV	30,31
DD2	1Q1 "	"	2,3,8,9,15,16, 23,29,30,35,36	25	22	"	"	"	"
DD3	1Q2 "	"	2,8,9,15,16,22, 23,29,30,35,36	6	3	"	"	"	"
DD4	1Q3 "	"	2,3,8,9,15,16, 22,29,30,35,36	26	23	"	"	"	"
DD5	1TC "	"	2,3,8,9,15,16, 22,29,30,35,36	24	23	"	"	"	"
DD6	2Q0 "	"	2,3,8,9,15,16, 22,23,29,35,36	34	30	"	"	"	"
DD7	2Q1 "	"	2,3,8,9,15,16, 22,23,30,35,36	14	29	"	"	"	"
DD8	2Q2 "	"	2,3,8,15,16,22, 23,29,30,35,36	33	9	"	"	"	"
DD9	2Q3 "	"	2,3,9,15,16,22, 23,29,30,35,36	13	8	"	"	"	"
DD10	2TC "	"	2,3,9,15,16,22, 23,29,30,35,36	31	8	"	"	"	"
DD11	3Q0 "	"	2,3,8,9,15,22, 23,29,30,35,36	19	16	"	"	"	"
DD12	3Q1 "	"	2,3,8,9,15,16, 22,23,29,30,35	39	36	"	"	"	"
DD13	3Q2 "	"	2,3,8,9,16,22, 23,29,30,35,36	18	15	"	"	"	"
DD14	3Q3 "	"	2,3,8,9,15,16, 22,23,29,30,36	38	35	"	"	"	"
DD15	3TC "	"	2,3[illegible]15,16, 22,2[illegible],30,36	40	35	"	"	"	"

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TABLE XXIII. Output crosstalk (high-level, low-level, and duration) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	D to pin	E to pin	Min	Max		
DD16	1Q0 Output crosstalk, duration	Open	3,8,9,15,16,22, 23,29,30,35,36	5	2	0	400	mV	30,32
DD17	1Q1 "	"	2,3,8,9,15,16, 23,29,30,35,36	25	22	"	"	"	"
DD18	1Q2 "	"	2,8,9,15,16,22, 23,29,30,35,36	6	3	"	"	"	"
DD19	1Q3 "	"	2,3,8,9,15,16, 22,29,30,35,36	26	23	"	"	"	"
DD20	1TC "	"	2,3,8,9,15,16, 22,29,30,35,36	24	23	"	"	"	"
DD21	2Q0 "	"	2,3,8,9,15,16, 22,23,29,35,36	34	30	"	"	"	"
DD22	2Q1 "	"	2,3,8,9,15,16, 22,23,30,35,36	14	29	"	"	"	"
DD23	2Q2 "	"	2,3,8,15,16,22, 23,29,30,35,36	33	9	"	"	"	"
DD24	2Q3 "	"	2,3,9,15,16,22, 23,29,30,35,36	13	8	"	"	"	"
DD25	2TC "	"	2,3,9,15,16,22, 23,29,30,35,36	31	8	"	"	"	"
DD26	3Q0 "	"	2,3,8,9,15,22, 23,29,30,35,36	19	16	"	"	"	"
DD27	3Q1 "	"	2,3,8,9,15,16, 22,23,29,30,35	39	36	"	"	"	"
DD28	3Q2 "	"	2,3,8,9,16,22, 23,29,30,35,36	18	15	"	"	"	"
DD29	3Q3 "	"	2,3,8,9,15,16, 22,23,29,30,36	38	35	"	"	"	"
DD30	3TC "	"	2,3,8,9,15,16, 22,23,29,30,36	40	35	"	"	"	"

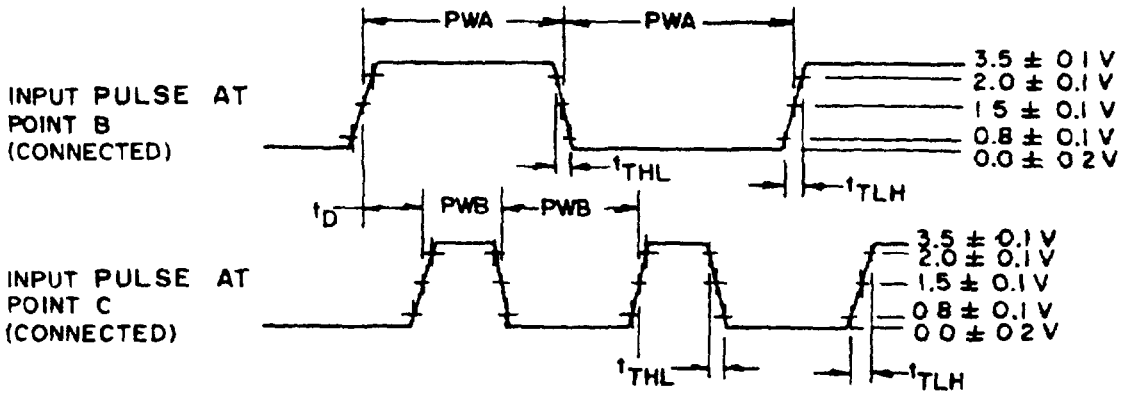
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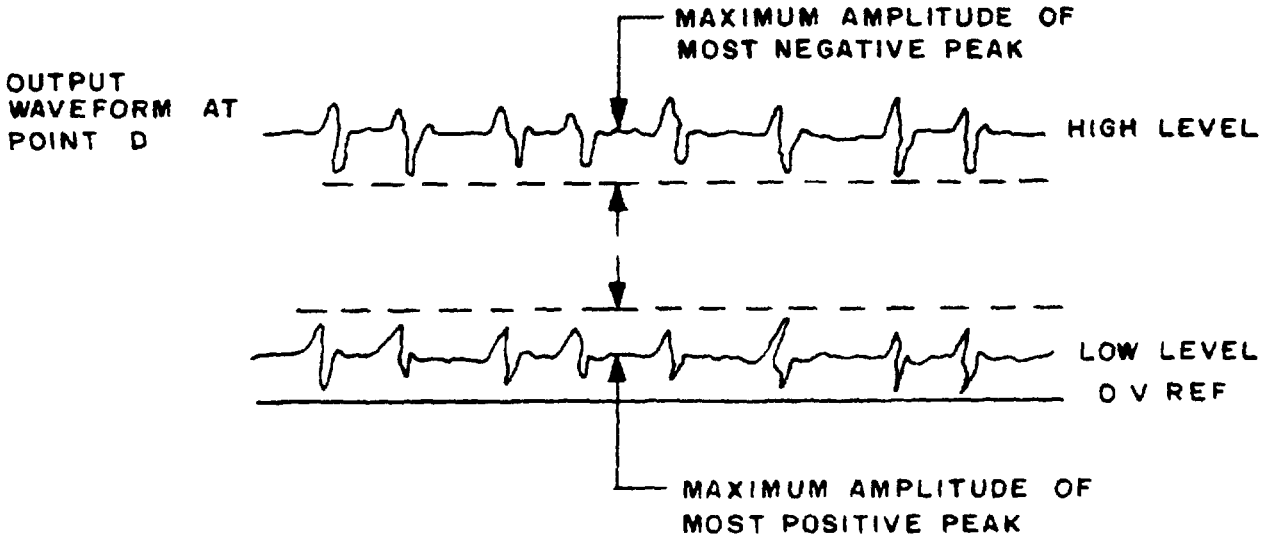


NOTES.
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as specified on figure 10.
3. Peak noise on V_{CC} shall be less than 400 mV.
4. Item J01-2 is operated in the double pulse mode.

FIGURE 30. Output crosstalk (high-level, low-level, and duration) test circuit for table XXIII.

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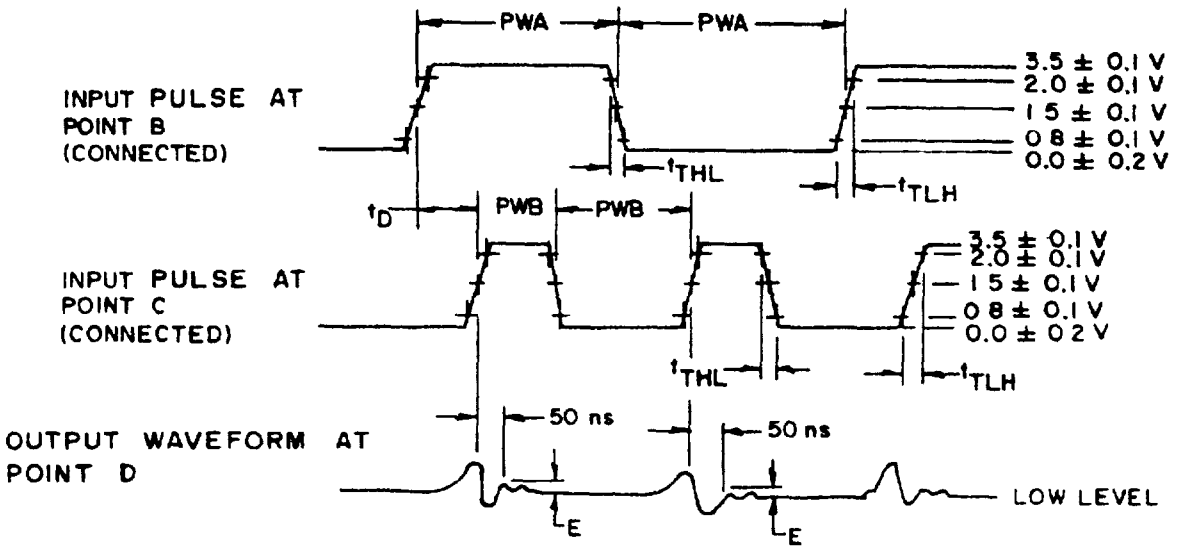




NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
PWB = 250 ±25 ns.
t_D = 125 ±25 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

* FIGURE 31. Output crosstalk (high-level and low-level) test circuit waveforms for table XXIII.

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NOTES

1. Input pulse parameters
   PWA = 500 ±50 ns.
   PWB = 250 ±25 ns.
   t_D = 125 ±25 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
2. E is the maximum positive amplitude after a settle time of 50 ns has elapsed from the time the CP made a positive transition.

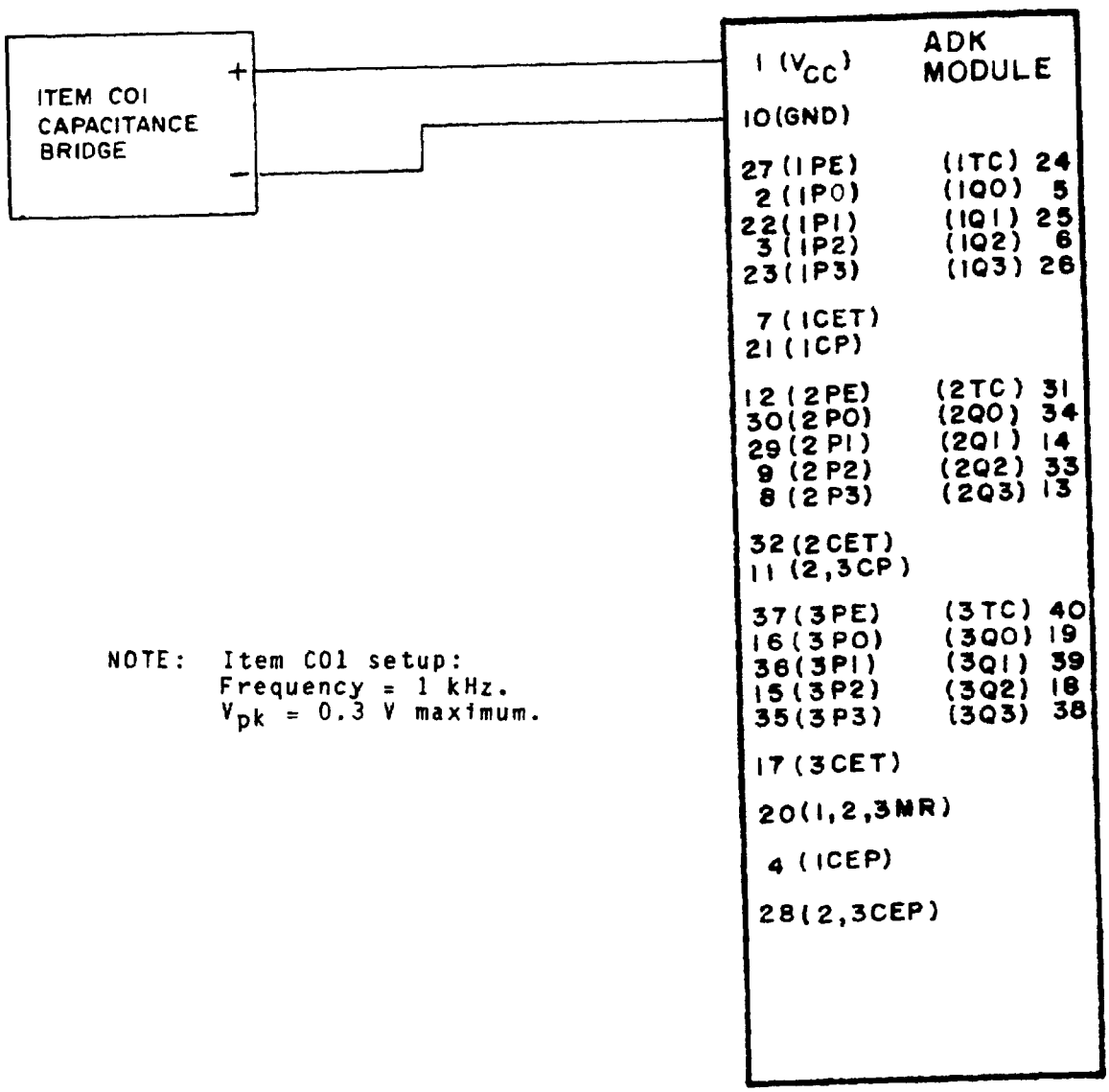
* FIGURE 32. Output crosstalk (duration) test circuit waveforms for table XXIII.

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TABLE XXIV. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
EE	Power supply filter capacitance	1.6	3.0	µF	33

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NOTE: Item C01 setup:
Frequency = 1 kHz.
V_{pk} = 0.3 V maximum.

* FIGURE 33. Power supply filter capacitance test circuit for table XXIV.

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5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

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